

Auditing Emergence in Quantum Error Correction: A Falsifiable Instantiation of Six Birds Theory

Ioannis Tsiokos 

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Abstract

Six Birds Theory (SBT) describes emergence as packaging: a stable macroscopic object arises when microscopic detail is repeatedly coarse-grained and restored to a canonical form. Its central methodological claim is that six primitives turn any such packaging structure into concrete computations that can be checked and can fail. We test this claim on quantum error correction, where the packaged object is a logical qubit, in an open, deterministic simulation of small repetition and surface codes. Theorem-level quantities are computed in exact rational arithmetic, and every prediction was registered before its scored run. Of the 25 predictions in the original build, 14 failed; all are reported. Four primitives nevertheless produced working audit objects. Extended with product features, the adequacy residual attains exactly the minimum mean-squared error with which any estimator can recover the logical observable of the three-qubit repetition code from its syndrome. The existence certificate rejects a decoder that is perfectly stable only because it collapses the logical qubit. Exact memory witnesses separate hidden-state noise from memoryless noise, and a protocol-trap control overturned our own hypothesis. Every registered prediction about the pricing primitive failed, and transport was not attempted. The centerpiece is a failed syndrome-only drift detector. We diagnosed it with the framework’s own exact machinery, corrected it, and re-registered it together with a prediction that it would still lose to the baseline on the original test; that prediction held, as did a large predicted win elsewhere. The corrected detector also cleared its registered bar under circuit-level noise, at the narrowest margin the test grid resolves. In a batch-decoded recalibration loop on one channel known to drift, it beat an offline blind schedule matched retrospectively to its number of recalibrations on each seed; whether it beats a schedule with five times its average number of recalibrations is unresolved. The error-correction mechanisms are not new; the contribution is the audit framing, the pre-registration discipline, and a falsification-and-repair loop carried out inside the framework. All code and results are open, and the nine default experiments regenerate byte-identically in the declared environment.

Keywords. quantum error correction; emergence; emergence calculus; Six Birds Theory; adequacy residual; existence certificate; pre-registered predictions; drift detection; decoder recalibration; reproducible simulation.

1 Introduction

Frameworks for emergence face a standard objection: they redescribe without adding knowledge. Call syndrome extraction “packaging” and decoding “lifting,” and you have new vocabulary for a logical qubit, not a new fact about it. Six Birds Theory (SBT) [69, 70] is a framework of this kind, and its foundational papers set the standard it has to meet. The framework is worth having only where it changes the *audit layer*: where its primitives compute something checkable about a domain, and where those computations can come out wrong. This paper tests that standard on quantum error correction (QEC), in the falsificationist sense [55, 46].

The test. SBT offers six audit primitives, all built on one *packaging operator*: an existence (closure) certificate, an adequacy residual, a predictive quotient that detects hidden memory, a protocol trap, a shadow-price budget audit, and transport. For each one we asked what it must become when the domain is a small stabilizer code under stochastic Pauli noise. We implemented five of the resulting objects in a deterministic simulation prototype. The sixth, transport, was scoped out before the build began and was never implemented. Quantities that carry theorem-level weight are computed in exact rational arithmetic. Before each scored run we froze quantitative predictions with numeric pass/fail bars: before implementation for experiments E1–E7, and from a pilot on disjoint seeds for the two circuit-level experiments E8–E9 (Section 4.1). A failed prediction is a result in its own right. We report it with its bar, its measured value and a diagnostic reading, and we never re-run it with tuned parameters.

The outcome. Figure 1 shows every registered prediction and its verdict. Of the original build’s 25 predictions, 14 failed as frozen. Four primitives nevertheless produced positively evidenced, machine-checkable audit objects:

- the *adequacy residual*, which measures how much of the logical information the scheduled checks leave uncovered;
- the *existence certificate*, a typed verdict on whether a stable logical layer exists at all;
- the *memory witness*, an exact certificate that histories which look identical now have different declared futures;
- the *protocol-trap* control, which ran correctly and overturned our own hypothesis that a test schedule was an artifact.

The fifth primitive, the pricing audit, runs exactly, but every one of its registered predictions failed. We report it as a negative.

Why publish a 56% failure rate? Because the failures are where the framework proved its worth, and leaving them in the file drawer [60] would hide that. The centerpiece of the paper (Section 7) follows one of them. A syndrome-only drift detector, the kind one could deploy, never reached its registered detection criterion (an alarm in at least 9 of 10 seeds) anywhere in the tested grid, up to $N = 16000$ shots. Instead of tuning it, we diagnosed it with the framework’s own exact machinery. A Walsh–Hadamard inversion recovers the full syndrome distribution in closed form, and with it we found two exact defects in the statistic, one mis-calibrated null, and one structural fact about the test itself: by construction, the frozen drift sits almost on a minimum-weight logical-failure path. Of the three scenarios we tested, this is the one where a syndrome-only method is at the greatest disadvantage relative to monitoring logical failures. We then re-registered a corrected detector with new frozen predictions. One of them predicted that the corrected detector would *still* not beat the baseline on the original scenario. That prediction held, and so did the prediction of a large win on a drift placed away from the logical operators. The correction carried over to circuit-level noise, where it cleared its registered bar at the narrowest margin the test grid can resolve. It also carried over to a closed decoder-recalibration loop on one channel known by construction to drift, with each batch of shots decoded after the recalibration decision for that batch. There, witness-triggered recalibration ended up about 0.0014 above a model-known oracle decoder in post-drift logical error rate. It beat, by 0.0185, a blind schedule that was given, retrospectively and offline, the same number of recalibrations on each seed. Whether it also beats a blind schedule with five times its average number of recalibrations is unresolved at ten seeds.

Original build: 25 registered predictions (E1–E6)

E1 coverage residual	P1.1	P1.2	P1.3	P1.4		2 held, 2 failed
E2 drift witness	P2.1	P2.2	P2.3	P2.4		1 held, 3 failed
E3 check selection	P3.1	P3.2	P3.3	P3.4		4 held
E4 existence certificate	P4.1	P4.2	P4.3	P4.4	P4.5	2 held, 3 failed
E5 decoder memory	P5.1	P5.2	P5.3	P5.4		2 held, 2 failed
E6 pricing	P6.1	P6.2	P6.3	P6.4		4 failed

Follow-up: 9 scored predictions and 5 descriptive reports (E7–E9)

E7 corrected witness	P7.1	P7.2	P7.3	P7.4	P7.5	P7.6	P7.7	2 held, 3 failed
E8 circuit level	P8.1	P8.2	P8.cost					2 held
E9 closed loop	P9.1	P9.2	P9.3	P9.4				2 held

held
 failed
 predicted failure to beat baseline; held
 descriptive, no bar

Figure 1. Every registered prediction and its verdict. Each row is one experiment; each cell is one registered entry, labeled by its identifier in the prediction ledger (Appendix B). The original build (E1–E6) registered 25 predictions, of which 11 held and 14 failed. The follow-up (E7–E9) registered 9 scored predictions (6 held, 3 failed) and 5 descriptive reports without a bar. The amber cell, P7.3, is a prediction that the corrected drift witness would *fail* to beat the baseline on the original test scenario; it held.

What we do not claim. None of the individual QEC mechanisms is new. Detector-correlation calibration is standard experimental practice [20, 26, 27]. Decoder reweighting under drift already exists [64, 35, 83, 4], and sequential change detection is seventy years old [52]. No hardware was involved. We make no asymptotic threshold statements. The residual certifies coverage only relative to a declared class of estimators (Section 3.3). The contribution is the test and its outcome. When SBT’s primitives were forced into contact with a technical domain under pre-registration, they produced checkable structure, genuine falsifications, and a repair loop that worked inside the framework rather than around it.

How to read this paper. No prior knowledge of SBT is assumed. Section 2 introduces the framework from scratch, in the finite setting the prototype uses. Section 3 maps each primitive to its QEC object and states the limits that bind every result. Section 4 describes the evidence rules. Sections 5–7 give the results: certificates checked against ground truth, the failed predictions as a taxonomy, and the diagnosis-and-repair arc. Sections 8–9 place the work and state what it does and does not establish. Section 10 lists the environment and commands that regenerate every number. The appendices hold the notation, the full prediction ledger, the control ledger, the deviations log, a formula summary, and the version history.

2 Six Birds Theory for readers who have never seen it

This section is self-contained. It introduces only the SBT machinery the prototype uses, in the finite, classical-probability setting where the prototype lives; the SBT papers develop the same objects in more generality [69, 70]. QEC readers may find it useful to keep the intended dictionary in mind throughout. The *carrier* will be the error state of a code, the *lens* will be syndrome extraction and decoding, and the *package* will be the logical qubit.

2.1 The core picture: packaging

SBT starts from three ingredients. A *carrier* is a set Z of microscopic states with some dynamics, here a Markov kernel P for one time step. A *lens* is a map f from carrier states to a much smaller set of macroscopic labels. A *prototype* assignment gives, for each macroscopic label x , a canonical carrier distribution u_x : the state the system treats as “the representative of x .”

From these, SBT builds the *packaging operator*

$$E_{\tau,f}(\mu) = U_f(Q_f(\mu P^\tau)), \quad (1)$$

read from right to left: evolve the carrier distribution μ for τ steps, push it through the lens (Q_f , marginalization onto macro labels), and lift each macro label back to its prototype (U_f) [69]. In words: *let the world run, look at it coarsely, and snap what you saw back to a canonical form*. On SBT’s account, something deserves to be called an emergent object when this round trip is nearly idempotent (packaging twice is almost the same as packaging once) and when the macro layer it stabilizes is nontrivial, with more than one distinguishable state surviving.

The QEC cycle is visible in Eq. (1). Noise acts (P^τ), a syndrome is extracted and decoded (Q_f), and recovery returns a canonical representative of the decoded logical class (U_f). We did not impose this as an analogy; it is what the framework produces once the carrier and the lens are chosen (Section 3). Two modelling choices are worth stating up front. Errors accumulate inside P , and recovery happens only in the packaging step, once per cycle, not after every step of P . For the hidden-state noise models, the prototypes start the hidden mode in its first state; this is a declared mathematical completion of the package, not a physical procedure for resetting leakage. What the framework adds is a list of things to *compute* about this loop. Each of the six primitives below is an audit: a quantity computable from the declared model whose value certifies something specific about the package, or exhibits a specific way it fails.

2.2 Primitive 1: the existence certificate

The first audit asks the bluntest question: *is there a stable macro object here at all?* SBT answers with a handful of numbers [69, 70, 79].

- The *idempotence defect* δ : how far E^2 is from E in total variation, maximized over initial carrier states. Small δ means packaging is self-consistent. (Maximizing over point masses suffices, by convexity.)
- The *retention error* ε : the worst case, over macro labels, of the probability that τ steps of dynamics carry a prototype out of its own macro cell. A theorem of the framework (T-IC-02 in [69]) guarantees $\delta \leq \varepsilon$. The reason is that one more cycle mixes prototype defects convexly, and lifting to prototypes cannot increase total-variation distance. A computed violation would therefore expose a bug in the implementation or an error in the theory.

- The *closure deficit* $CD_\tau = I(X_t; Y_{t+\tau} | Y_t)$, with $Y = f(X)$: how much the hidden micro state tells you about the macro label τ steps ahead, beyond what the current macro label already tells you. The mutual information is taken under the chain’s stationary law; for absorbing models, where that law degenerates, a declared finite-horizon weighting takes its place (Appendix D). $CD_\tau = 0$ means that, on the support of the weighting law, the current macro label is sufficient for predicting the later one. This is the lumpability condition of classical Markov-chain theory [39, 8] at horizon τ , expressed as an information-theoretic closure measure of the kind also used outside SBT [54]; SBT’s version is in [70, 73, 68].
- The *route mismatch* RM_τ : whether the micro states inside one macro cell agree on where they send the macro label next. It is the weighted ℓ_1 spread of each micro state’s coarse transition profile around its cell’s average, summed over cells [79]. It vanishes exactly when the coarse transition profiles agree within each cell on the support of the weighting law. Full lumpability of the chain additionally requires that law to have full support.
- A *multiplicity* count: how many macro labels have stable prototypes. This guards against a degenerate “certificate” obtained by collapsing everything to one state. A decoder that maps every error to “fine” has perfect idempotence and no object.

Frozen thresholds turn these numbers into a typed status: `certified`, `degrading`, `non_closed`, or `trivialized`. The status matters more than any single number. It separates “the object exists” from “the object is dissolving,” from “the macro layer leaks information from a hidden coordinate,” and from “you defined the object away.”

2.3 Primitive 2: the adequacy residual

The second audit asks: *do the observables you actually schedule cover the thing you care about?* Fix a family L of *native probes*, the things you measure routinely, and a family D of *dissolving probes*, the things whose values define the object but which you cannot measure directly without destroying it. The *adequacy residual* is the Schur complement

$$\Xi(D | L) = K_{DD} - K_{DL} K_{LL}^+ K_{LD}, \quad (2)$$

where the K blocks are covariances among the probe observables and $^+$ is the Moore–Penrose pseudoinverse [78]. Because a joint covariance is a Gram matrix, its off-diagonal block always lies in the range of K_{LL} , so the formula is well defined even when K_{LL} is singular. $\Xi(D | L)$ is the residual covariance of the best *affine* estimator of D from L : linear in the observables, with an intercept. In general it is *not* the conditional covariance of D given L (the two coincide, for example, in the Gaussian case), so we call it a *residual* throughout. It is the part of the object’s behaviour that the scheduled measurements cannot see, priced in variance. Three properties give the primitive its teeth.

- *The blind-spot witness.* If $\Xi(D | L)$ exceeds a declared allowance Ω , so that $\lambda_{\max}(\Xi - \Omega) > 0$, then the top eigenvector z of $\Xi - \Omega$ *names a direction*: the combination of dissolving observables most invisible to the current schedule [78].
- *The chain rule.* Adding a candidate probe family M lowers the residual by an exactly computable amount, $\Xi(D | L \cup M) = \Xi(D | L) - \text{discharge}(M | L, D)$. The discharge is obtained by first removing from M what L already explains, then projecting onto what is left. Marginal coverage is therefore an exact, decomposable currency, and each step of a greedy probe selection is exactly computable [78].

- *Saturation.* A duplicated probe has exactly zero discharge. This is a built-in null: an implementation in which redundancy appears to buy coverage is wrong [78].

One limit applies to everything the residual says. $\Xi = 0$ certifies that D is, almost surely, an affine function of the declared probe family. A nonzero Ξ certifies a gap in at least one direction *for that class*; positive definiteness, $\Xi \succ 0$, is the stronger statement that every direction has a gap. Neither says anything about nonlinear decoders outside the declared family. The framework’s own account of nonlinearity is the *degree ladder*: enlarge the probe family with product features (degree 2, degree 3, ...) and watch the residual contract through the same chain rule [78, 68].

2.4 Primitive 3: predictive quotients and the memory witness

The third audit asks: *does acting well require remembering more than you can currently see?* Its conceptual counterpart outside SBT is the memory term of projection-operator reductions, which appears when a coarse-grained description fails to close [11].

Take a declared finite catalog of *histories*, each a probability distribution over carrier states encoding “what might be true now, given what was observed.” Two histories are *currently equivalent* if every currently measurable event has the same probability under both. They are *predictively equivalent* if every declared future observation, under every declared continuation of the dynamics, has the same probability under both. Partitioning the catalog in the two ways gives the *current quotient* Q and the *predictive quotient* M [72]. For M to refine Q , predictive equivalence must imply current equivalence. A finite catalog of future tests does not guarantee this on its own; including the identity continuation together with the current tests does. In this prototype the hidden-state catalogs include them, and refinement is verified directly for every catalog. Predictive equivalence of histories is also the construction behind causal states and ε -machines in computational mechanics [14, 62], and behind predictive state representations [45]. SBT restricts it to a declared finite catalog so that it can be computed exactly.

A *memory witness* is a pair of histories that are currently equivalent but predictively distinct: the present looks identical, the futures differ. Three numbers summarize the witnesses. The *witness count* is the number of such pairs. *MaxFiber* is the largest number of predictive classes hiding inside one current class. The worst-case predictive gap $\Delta^{\max}$ is an exact rational: the largest difference, over witness pairs and declared future events, in the probability of that event. Its operational meaning is a lower bound. A predictor that treats the two histories of a witness pair as one state must, on the event that separates them by g , be off by at least $g/2$ for one of them: a single number cannot be closer than half their separation to both probabilities, and the midpoint attains this. A predictor that uses only current information treats every currently equivalent pair as one state, including the pair that attains $\Delta^{\max}$, so its worst error on the declared events is at least $\Delta^{\max}/2$; the factor $1/2$ cannot be improved. These numbers certify *that* the declared futures differ and *by how much*. Whether the difference is worth anything to a decoder is a separate, measured question (Section 5.3).

Two constructive companions complete the audit. *Currentization* searches a declared candidate list for the smallest set of added observables that dissolves every witness; this exposes what the hidden state is. The *minimal machine* is the smallest automaton that reproduces a declared set of future predictions [72, 71]. In this prototype it is computed for a declared, quantized belief automaton and its next-syndrome predictions (Section 5.3).

2.5 Primitive 4: the protocol trap

The fourth audit is a control on the third. Apparent memory can be an artifact of an external schedule. If an experimenter alternates conditions, memory of the current phase can masquerade as memory intrinsic to the system. The protocol trap prescribes *internalization*: rebuild the model with the schedule as an explicit, autonomously evolving state coordinate, and recompute the witnesses. If they vanish, the memory was a scheduling artifact (`artifact_trap`); if they survive, it is genuine [75, 69]. This is a falsification instrument aimed at the framework’s own favourite conclusion, which is why we implemented it. As Section 5.3 reports, it overturned *our* expectation.

2.6 Primitive 5: prices and slack

The fifth audit treats coverage as an economy. Probes carry costs, and a budget b limits what can be scheduled. The value function $V(b)$ is the most residual that budget b can discharge, and the *shadow price* $\lambda(b) = V(b+1) - V(b)$ is the value of one more unit of budget [77]. Because the feasible probe sets grow with b , V is nondecreasing. Here λ is a discrete finite difference, not the multiplier of a continuous optimization, and V need not be concave. The audit makes four kinds of claim. The shadow price prices the marginal check. A *slack point* b^* , the smallest budget beyond which λ stays below a declared tolerance, certifies over-provisioning: above b^* , each unit of budget removed changes the residual by at most the tolerance. Below b^* the budget should matter, and a consequence test asks whether removing the last unit before b^* shifts the decoder’s sampled logical error rate detectably. Finally, optimizing against a *proxy* cost vector, with costs shuffled among candidates, should be visibly worse; this null detects “economics” that is merely decorative. In this prototype the pricing machinery is exact and runs correctly, and every one of its registered predictions failed at the frozen tolerances. Section 5.4 reports this negative in full.

2.7 Primitive 6: transport (not tested here)

The sixth primitive, transport (or functoriality), would supply certified bridges showing that a package deforms coherently across related carriers, with composable defect bounds [74]. It was deferred before this build began and was never implemented. It appears in this paper only in the instantiation map (Figure 2), greyed out, so that no reader comes away thinking all six primitives were demonstrated.

2.8 The methodological spine

Two further SBT commitments shaped everything below. They are about method rather than mathematics. The first is *exact-finite evidence*: theorem-grade quantities are computed on declared finite models in exact rational arithmetic, so that a certificate is a computation, not an estimate [70]. The second is the *null-battery discipline*: every positive mechanism ships with the controls that would expose it as fake, such as saturation nulls, trivialization guards, proxy failures and protocol traps [73]. The prototype implements the first commitment on its exact engines (Section 4.3 says precisely which computations are exact) and the second in full. Section 4 describes the resulting evidence rules.

3 Instantiation on quantum error correction

3.1 The domain objects

Codes. The prototype uses two code families in four instances, all declared and frozen before implementation: REP(3), REP(5) and SURF(3) as the primary instances, and SURF(5) only for scaling spot-checks. REP(d) is the classical repetition code with checks $Z_i Z_{i+1}$. It protects against bit flips only; read as a quantum stabilizer code, its phase-flip distance is 1. Experiments on REP codes therefore track only the protected logical observable. They use bit-flip noise, except the hidden-state studies on REP(3), whose depolarizing noise the code sees only through its bit-flip component. SURF(d) is the rotated surface code [40, 6, 16]; SURF(3) has 9 data qubits and 8 checks. Code conventions follow the standard rotated layout [19, 67], and the declared distances of SURF(3) and SURF(5) are verified in the test suite by exhaustive search for low-weight undetected logical errors.

Noise. Five noise models are used, all stochastic Pauli noise at the code-capacity level unless stated otherwise.

- N1: independent bit flips with probability p .
- N2: independent depolarizing noise with probability p .
- N3: N2 plus an *unmodeled* correlated ZZ error on one declared qubit pair with probability q . This is the drift model: the decoder’s declared model still believes N2.
- N4: a hidden two-state Markov mode that switches the depolarizing rate between p_0 and $3p_0$. The mode is not observable.
- N5: a latching proxy for leakage. One designated qubit irreversibly becomes fully depolarizing, at rate r per round.

N4 is a telegraph-type rate fluctuation of the kind that two-level-system defects produce in superconducting qubits [41], and temporally correlated noise of this kind is known to degrade surface-code memories [37]. N5 is deliberately crude; real leakage in stabilizer circuits is more structured [22, 47]. Sections 7.4–7.5 also use circuit-level noise: Stim-generated rotated-surface-code memory circuits [24] with depolarizing and measurement-flip errors at every location, decoded by minimum-weight perfect matching [16, 17] with PyMatching [31, 32].

The moment engine. Everything exact rests on one identity. A Pauli error and a check or logical observable are both vectors in $\mathbb{F}_2^{2n}$ [29]. The detection bit is their symplectic pairing $\langle a, e \rangle$, and the corresponding ± 1 observable is $\sigma_a(e) = (-1)^{\langle a, e \rangle}$. Because $\sigma_a \sigma_b = \sigma_{a \oplus b}$, every covariance block in Eq. (2) reduces to means of such observables. Under independent qubit noise, each mean $\mathbb{E}[\sigma_a]$ is a product of per-qubit rational factors. N3’s correlated injection, being independent of the rest, multiplies this product by its own factor, and hidden-mode models are evaluated one mode at a time. Every covariance block is therefore exactly computable without sampling. The same identity later makes the diagnosis of Section 7.2 possible. Means of XOR-combinations of checks *are* the Walsh–Hadamard coefficients of the joint syndrome distribution [51], so the whole 2^8 -outcome syndrome distribution of SURF(3) can be recovered exactly from the moment engine. The same Boolean-Fourier fact underlies identifiability results for learning Pauli channels from syndrome statistics [82].

3.2 The instantiation map

Table 1, drawn as Figure 2, is the frame of the paper. For each SBT primitive it lists the QEC object it becomes, the experiments that test it, and the evidential outcome. Four primitives are positively evidenced, one is negative, and one was deferred.

SBT primitive	QEC instantiation	Experiments	Evidential outcome
Packaging operator $E = U_f \circ Q_f \circ P^r$	the QEC cycle: noise $\rightarrow$ syndrome $\rightarrow$ recovery	all	substrate for every audit
Existence certificate ($\delta, \varepsilon, CD_\tau, RM$, multiplicity)	typed status of the logical qubit as an object	E4 (+E1)	positive (§5.2)
Adequacy residual $\Xi(D \mid L)$, witness, chain rule	syndrome-coverage audit over native and degree-2 checks	E1, E2, E3, E7–E9	positive (§5.1, §7)
Predictive quotient / memory witness ($Q, M, \Delta^{\max}$)	decoder-memory certificate; minimal machine for a quantized predictor	E5	positive (§5.3)
Protocol trap / internalization	schedule artifact vs. genuine memory	E5	control ran; artifact hypothesis falsified (§5.3)
Shadow price / slack (V, λ, b^*)	economics of the check budget	E6	machinery exact; all registered predictions failed (§5.4)
Transport / functoriality	certified deformation bridges	—	deferred; never implemented

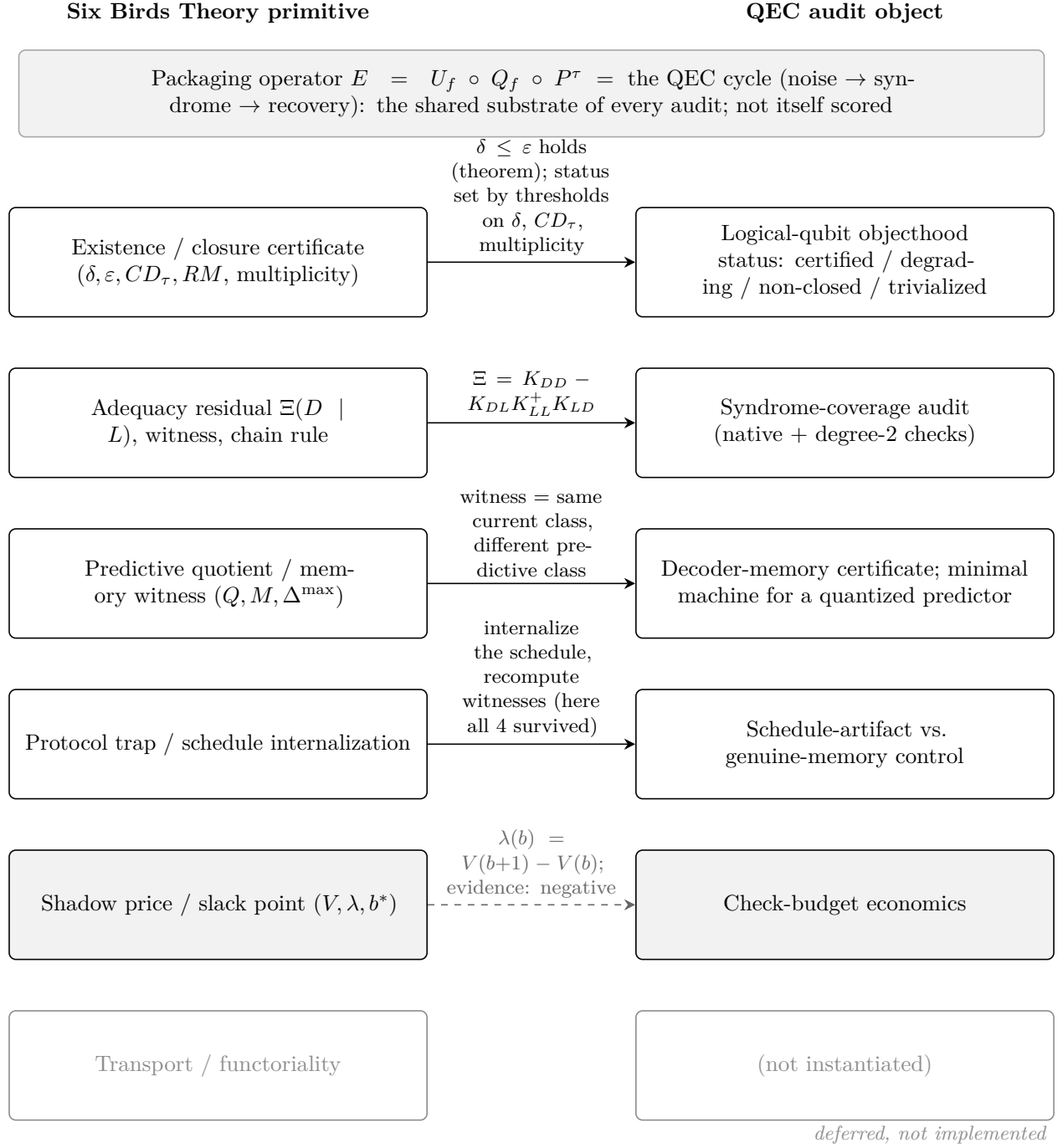
Table 1. The instantiation map. The last column gives the outcome for each primitive: four positively evidenced, pricing negative, transport deferred.

3.3 Scope

The following limits were fixed in the design documents before implementation. They bind every result in this paper.

1. *No quantum hardware.* All evidence is classical simulation of stochastic Pauli and classical Markov models, plus Stim circuit-level sampling.
2. *The residual is relative to an estimator class.* The prototype’s Ξ is the residual covariance of the best affine predictor of the logical ± 1 observables from the scheduled-check ± 1 observables, optionally augmented with degree-2 products. $\Xi = 0$ certifies exact affine coverage over the declared features. A nonzero Ξ certifies, and prices, a coverage gap for that class. It does *not* assert that no nonlinear decoder covers the gap; the degree ladder shows how the residual contracts as the feature degree grows.
3. *No Born-rule, Bell or coherent-dynamics claims.* Stochastic Pauli noise is a standard, declared restriction. It leaves out coherent errors [7] and quantum-bath non-Markovian dynamics [66]. The temporal correlations the prototype does model (N4, N5) are classical hidden-state correlations layered on stochastic Pauli noise. SBT’s own paper on quantum theory works at exactly this packaging level [76].
4. *No asymptotic threshold claims.* Every result concerns a finite carrier.

5. *Certificates are computed, not fitted.* Residuals and quotients are computed exactly on declared finite models, as are the idempotence defect, retention error and multiplicity of the existence certificate for the REP(3) packages under bit-flip noise. The remaining existence-certificate quantities (closure deficits and route mismatches everywhere, and all components for SURF(3) and the hidden-state models) are computed numerically on the same declared models, without sampling or fitting. Fitted macro models are used only as diagnostics. Sampled outcomes, including the E5 payoff study of predictors trained on part of a trajectory, are reported as measurements, never as certificates.
6. *The audits are independent.* Stability does not establish nontrivial storage: a trivialized decoder is perfectly stable, which is why multiplicity is a separate check. Coverage and existence do not imply one another, and neither do memory witnesses and drift. Each audit is certified or falsified on its own, and outcomes are never summed across audits.



solid arrow = positively evidenced audit object (4); dashed = machinery exact, every registered prediction failed (1); grey = deferred (1); top band = shared substrate, not scored

Figure 2. The SBT-to-QEC instantiation map. The packaging operator is the shared substrate of every audit and is not scored itself. Solid arrows: the four primitives that produced positively evidenced audit objects. Dashed arrow: the pricing primitive, implemented exactly, whose registered predictions all failed. Greyed row: transport, scoped out before the build began and never implemented.

4 Methods: the evidence rules

The prototype’s credibility rests on process, so we describe the process with the same care as the mathematics.

4.1 Registered predictions

Every scored prediction was written down with a numeric pass/fail bar in a frozen design document. Descriptive reports, which have no bar, were declared alongside. For E1–E7 this happened *before* the experiment was implemented. For E8–E9 the simulation infrastructure had to exist before a pilot could run, so the bars were frozen from a pilot on disjoint seeds before the scored run (below). This is pre-registration in the sense of [50], applied to a simulation study. A failed prediction is reported as REGISTERED-NEGATIVE with its frozen bar and measured value. It is never deleted, softened, or re-run with tuned parameters; a changed re-run would be a new experiment, labeled as such. The original build froze 25 predictions across six experiments (E1–E6). The follow-up of Section 7 froze 7 more for E7 (5 scored, 2 descriptive), 3 for E8 (2 scored, 1 descriptive) and 4 for E9 (2 scored, 2 descriptive). The registration records are the dated design documents and the deviations file. The public repository was released as a single commit, so the chronology rests on those records rather than on version-control history.

For the circuit-level experiments E8 and E9 there is no closed-form moment engine, so exact pre-derivation was replaced by a declared *pilot-then-freeze* protocol. Effect sizes were estimated on a pilot seed range disjoint from the scored seeds. Numeric bars were then frozen in the design addendum and in the runners’ registered-prediction code before the scored run, and the pilot-to-freeze step was itself logged.

4.2 Evidential grades

Every scientific statement in this paper carries exactly one of six grades, fixed in the design documents before any experiment ran.

- THEOREM-ANCHORED: an instance of a cited SBT theorem, verified computationally. This is the strongest grade the prototype can issue.
- EXACT-FINITE: an exact rational computation on a declared finite model.
- MEASURED: seeded sampling, reported with its uncertainty where one is computed.
- REGISTERED-POSITIVE and REGISTERED-NEGATIVE: a frozen prediction that passed, or failed.
- INTERPRETATION: a connecting reading. It carries no evidential weight.

Infrastructure statements, such as test counts and hashes, carry no grade. A sentence never mixes grades; where a result and its reading belong together, they are split into separate sentences.

4.3 What is exact, and what is not

The exact engines use rational arithmetic (Python `Fraction`) end to end, and the test suite checks them against brute-force enumeration. They are: the moment engine’s exact branch; the residual, discharge and chain-rule construction, including the reported chain-rule discrepancy; the whole predictive-quotient engine; exhaustive pricing enumeration and greedy check selection, including the stopping rule and the budget arithmetic; and the comparisons behind the saturation and exact-MMSE verdicts. The pseudoinverse is verified against all four Penrose identities as rational equalities. Some quantities are floating-point by design. These are the eigenvectors of the blind-spot witness (extracted from exactly computed matrices); stationary distributions from generic eigensolvers (where a model’s stationary law is uniform, this is checked exactly) and the finite-horizon weights

used for absorbing models; the existence-certificate computations on SURF(3) and on the hidden-state models; all logarithms, and hence every KL divergence, closure deficit and log-likelihood; all sampled quantities; and the circuit-level modules of Sections 7.4–7.5, which have no exact moment engine. The repository’s mathematics audit lists these boundaries file by file.

4.4 The null battery

Nine standing controls, the framework’s “null battery” [73], were implemented alongside the positive mechanisms.

1. The protocol trap (Section 2.5).
2. Same-family saturation: duplicated checks must show exactly zero discharge.
3. Null-model calibration: the false-positive rate of every witness under the declared model.
4. Proxy-cost failure.
5. Slack-regime checks.
6. No silent caps: any truncation of a candidate family is logged with exact counts.
7. The trivialization guard: a decoder that collapses the logical layer must be classified **trivialized**, not **certified**.
8. Same-support discipline: declared catalogs, no post-hoc edits.
9. A raw-history comparator: a seeded lookup predictor given the last three syndromes (current one included), trained on the first half of each trajectory and scored on the second, as a secondary baseline for the memory-payoff claims.

Appendix C gives the outcome of each.

4.5 Determinism and reproduction

Every run is seeded and configured from committed JSON. Identical inputs produce byte-identical result files, with floats serialized at full precision and keys sorted. Each experiment directory holds its configuration, its results, its figures together with the exact CSV data behind them, an environment record, and a SHA-256 manifest of every result and data file (the manifest and the environment record are outside the hash set). A single command regenerates all nine experiments and verifies their manifests. Two consecutive runs in the declared environment produced byte-identical files, and the test suite passes 209 tests. This is computational reproducibility in the sense in which the executable computation, not a description of it, is the deliverable [61, 53]. Hash integrity, reproducibility and mathematical validity are separate checks; the tests and the audit address the last. Section 10 gives the exact environment and commands.

4.6 Deviations log

Every divergence between the frozen designs and the running code was recorded, dated and explained in a public deviations file rather than silently absorbed. The log has 11 entries (Appendix D). Eight date from the build. Several of those are self-caught correctness fixes made *before* results were frozen, including two rounds of pilot-driven repair to the closed-loop witness policy of Section 7.5. One is a correction to a cross-check estimator whose log entry was added only after an external adversarial review found the fix documented in prose but missing from the log. The closed-loop entry also records a budget-matching correction made after that experiment’s first frozen run, which was then re-run and re-frozen. The remaining three entries record an audit of the mathematics and executable certificates (Appendix D). No entry changed a frozen bar. The deviations file is evidence for the process, not an embarrassment to minimize; that is why it is an appendix and not a footnote.

5 Results I: certificates on ground truth

This section reports the four audits on declared, exactly known models. This is where the framework’s claims are sharpest, because everything it certifies can be checked against ground truth. Experiments are labeled E1–E6 as in the repository. Registered verdicts appear here whether they passed or failed; Section 6 then looks at the failures as a group.

5.1 Coverage: the residual computes the right object (E1, E3)

Theorem checks. Two theorems of the framework were verified on every model where they apply (THEOREM-ANCHORED). The chain-rule identity $\Xi(D \mid L \cup M) = \Xi(D \mid L) - \text{discharge}(M \mid L, D)$ holds with a discrepancy of exactly zero in rational arithmetic on the tested models, and $\delta \leq \varepsilon$ holds on every model built. Section 10 names the tests that check them.

Residuals and ablations. With every check scheduled, the residual traces are $\text{tr } \Xi = 0.0837$ for REP(3), 0.0740 for REP(5) and 0.1754 for SURF(3) (EXACT-FINITE). Dropping any single check strictly increases the residual, and on REP(5) dropping an end check differs from dropping a middle one, as registered (REGISTERED-POSITIVE). On average an end-check drop costs more: the mean residual increase is 0.0593 for end checks and 0.0204 for middle checks (EXACT-FINITE). Duplicated checks show exactly zero discharge, both in E1 and in E3’s candidate form (REGISTERED-POSITIVE). The saturation null holds exactly, as the theory requires.

The cleanest result: the degree ladder reaches the exact optimum. Take REP(3) under bit-flip noise and extend its two native checks to the complete degree-2 family by adding their product. The residual trace drops from 0.083680 to 0.027575 (Figure 3). The second number equals, exactly, the minimum mean-squared error with which *any* function of the syndrome can estimate the logical ± 1 observable, which we obtain independently by enumerating the error distribution. Both are the rational number 47291/1715000, and the experiment confirms their equality as an equality of fractions (EXACT-FINITE). The registered prediction that the top rung equals the exact minimum mean-squared error therefore held (REGISTERED-POSITIVE). The equality is exact for a simple reason. REP(3) has four possible syndromes, and the intercept, the two syndrome bits and their product span every function on four points. The best affine estimator over that family is therefore the best estimator of any kind, namely the conditional mean. Note what this quantity is: a squared-error estimate of the logical observable, not the success probability of a decoder that must output one bit. This is our sharpest instance of “the audit computes the right object”: a coverage certificate derived from SBT’s chain rule lands, digit for digit as a fraction, on a quantity a QEC practitioner would compute by brute force. We use the word “optimal” for this result because it attains the enumerated optimum. Elsewhere in the paper the word appears only in the standard names of mathematical objects, such as the coverage-optimal map A^* , or as a by-construction property of the first greedy step, never as a performance claim.

Greedy selection matches or beats the tested orders at every budget. On SURF(3), greedy check selection by the chain rule lowers the residual from 0.4345 with no checks to 0.1754 with all of them. At *every* intermediate budget its residual is at or below that of every tested baseline: lexicographic order and 10 seeded random orders (Figure 4; REGISTERED-POSITIVE). This is a registered comparison against those baselines, not a general optimality statement. Only the first greedy step is optimal by construction. Greedy selection carries the classical $(1 - 1/e)$ guarantee only for submodular objectives [48, 43], and variance reduction by a set of regressors is not submodular

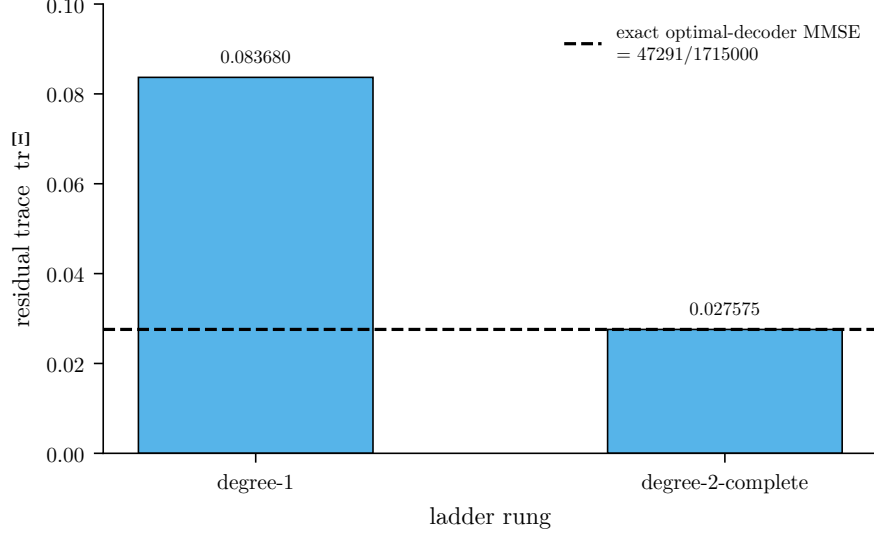


Figure 3. Degree-ladder contraction on REP(3) under bit-flip noise (E3). The residual of the complete degree-2 family equals the enumerated minimum mean-squared error of the logical observable exactly: both are 47291/1715000 (EXACT-FINITE). Intercept, the two syndrome bits and their product span every function of the REP(3) syndrome, so the best affine estimator over this family is the conditional mean itself. The quantity is a squared-error estimate, not a binary decoding error rate.

in general [15], so we claim no such guarantee. Given a candidate list padded with duplicates, the greedy selector also stops before exhausting it rather than spending budget on checks that provably add nothing (REGISTERED-POSITIVE).

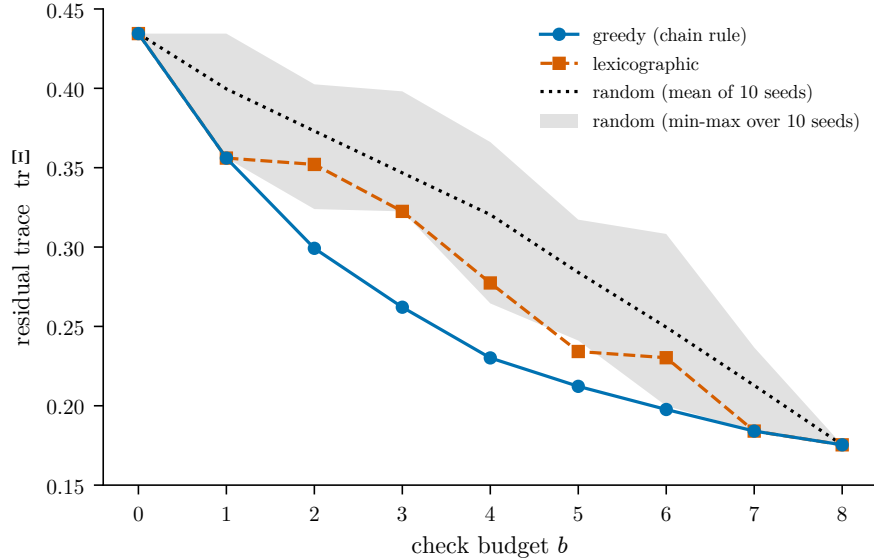


Figure 4. Residual trace versus check budget under chain-rule greedy selection on SURF(3) (E3). Greedy is at or below the lexicographic order and every one of 10 seeded random orders at every budget (REGISTERED-POSITIVE). This compares against the tested baselines only; just the $b = 1$ step is optimal by construction.

5.2 Existence: the certificate does classification work (E4)

The typed existence certificate was computed on ground-truth Markov models of the error-correction cycle. For the REP(3) packages under bit-flip noise in Table 2, its idempotence defect, retention error and multiplicity are exact rational computations (EXACT-FINITE); closure deficits, and the certificates of the hidden-state models below, are evaluated numerically. The baseline package is REP(3) under bit-flip noise with the standard decoder. It certifies, with $\delta = 0.007144875$, $\varepsilon = 0.00725$ (so the theorem bound $\delta \leq \varepsilon$ is nearly tight), multiplicity 2, and status **certified**. A deliberately broken decoder that maps everything to one logical state is caught by the multiplicity guard. It has $\delta = 0$ *exactly*, perfect idempotence, but multiplicity 1, so its status is **trivialized** (REGISTERED-POSITIVE for the guard). The pair captures the certificate idea in one contrast. A small idempotence defect alone would have certified the broken decoder; the typed certificate refuses it.

Sweeping the physical error rate shows the defect growing (EXACT-FINITE): $\delta = 0.000298$, 0.007145, 0.026432 and 0.082368 at $p = 1/100$, $1/20$, $1/10$ and $1/5$. The sweep labels a package **degrading** once δ exceeds 0.05, which first happens at $p = 1/5$. That label uses the defect alone. The full classifier would return **trivialized** at $p = 1/5$, because no prototype is stable there (multiplicity 0, $\varepsilon = 0.104$) and multiplicity takes priority over the defect threshold (EXACT-FINITE). Table 2 collects the classification.

package	δ	ε	mult.	status
REP(3) + N1, standard decoder, $p = 1/20$	0.007144875	0.00725	2	certified
same, deliberately broken decoder	0 (exact)	0	1	trivialized
same, $p = 1/5$ (sweep rule)	0.082368	0.104	0	degrading [†]

Table 2. The existence certificate doing classification work on ground truth (EXACT-FINITE). The broken decoder has *perfect* idempotence; the multiplicity guard, not the defect, is what catches it. The remaining status, **non_closed**, is set by a threshold on the closure deficit. Its registered demonstration on the hidden-mode model failed (REGISTERED-NEGATIVE; Appendix B, P4.2): the zero-deficit clause holds on the syndrome lens, not on the frozen decoded lens. [†]The sweep assigns its label from the defect threshold alone; with multiplicity 0 the full classifier would return **trivialized** at $p = 1/5$.

Two further components behaved as designed on hidden-state models. First, the closure deficit tracks an out-of-sample predictability proxy Δ_{pred} : the held-out improvement in next-label log-likelihood when a predictor of the macro label sees the previous label as well as the current one. Across the declared table of models and horizons, the Pearson correlation is $r = 0.936$ with the latching model N5 included (MEASURED). The registered bar $r \geq 0.9$ held (REGISTERED-POSITIVE). Two qualifications matter. Without N5 the correlation is $r = -0.171$, so the registered correlation is carried by N5 and does not hold on the remaining models. And the proxy is not the closure deficit itself. It measures a one-step macro-history gain in a long simulated run, started from the stationary law for nonabsorbing models and from the declared finite-horizon weighting for N5; it repeats the same value across horizons for a given model, and for N5 the long run can wash out the finite-horizon law that defines CD_τ ; the coefficient describes the declared table, not a sample of independent models. Second, the route-mismatch diagnostic shows the intended leakage signature on N5: the decoded label stays channel-accurate (0.75) while RM grows severalfold with the horizon, peaking at $7.28\times$ the baseline. It saturates below the frozen $10\times$ bar, a REGISTERED-NEGATIVE discussed in Section 6.

5.3 Memory: exact witnesses, a measured payoff, and a control that overturned our hypothesis (E5)

Witnesses separate the noise models exactly. The predictive-quotient audit was run on REP(3) packages under three noise models (EXACT-FINITE). Under memoryless noise (N1) it finds *no* memory witnesses: $|Q| = |M| = 4$ and $\Delta^{\max} = 0$. Under hidden-mode noise (N4) it finds 4 witnesses, MaxFiber 2, and $\Delta^{\max} = 112329952/1318359375 \approx 0.0852$. Under latching noise (N5) it finds 4 witnesses, MaxFiber 2, $|M| = 8$ predictive classes over $|Q| = 4$ current classes, and $\Delta^{\max} = 539/1250 = 0.4312$. Every package exports a well-defined map from predictive to current classes, confirming that M refines Q (EXACT-FINITE). These are certificates in the strict sense. Within the declared catalog of histories, events and continuations, they say exactly which currently indistinguishable histories assign different probabilities to some declared future event (on N4 and N5), and that no such pair exists on N1. Any predictor that uses only current information must therefore err by at least $\Delta^{\max}/2$ on some declared event: at least 0.0426 on N4 and $539/2500 = 0.2156$ on N5 (EXACT-FINITE). Whether a decoder can turn this difference into performance is the separate, measured question below.

Currentization names the hidden coordinate. Adding the mode bit as an observable dissolves every witness with a single added observable; no set of pure check bits does (REGISTERED-POSITIVE). The audit therefore does more than detect hidden memory. It identifies *which* added observable would expose it.

The protocol trap overturned our hypothesis. We built a two-phase alternating noise schedule as the intended “schedule artifact.” We registered the prediction that internalizing the schedule (adding the phase as an autonomous state coordinate) would dissolve its witnesses and classify it `artifact_trap`. That prediction *failed*: internalization preserved all 4 witnesses, and the package classifies as `genuine_memory_after_internalization` (EXACT-FINITE for the classification). The registered artifact hypothesis therefore failed (REGISTERED-NEGATIVE). Our reading (INTERPRETATION): a deterministic alternation between p_0 and $3p_0$ carries real predictive content, because the current phase predicts next-round statistics, so the trap correctly declined to call it an artifact. This is easy to misreport, so we state it plainly. The control mechanism worked, and what it falsified was *our* registered hypothesis. It is a negative result for that hypothesis, not a demonstration of the trap catching an artifact.

The minimal machine. The registered decoder machine is a declared, quantized belief automaton with 8 nodes: four syndrome values times two levels of a coarsened belief about the hidden mode. It is a packaged approximation, not the exact filter of the hidden Markov model. Exact partition refinement finds the smallest automaton with the same next-syndrome predictions. It has 2 predictive states, one for each belief level, and only 1 of them is reachable from the frozen initial prototype (EXACT-FINITE). At the frozen defaults the declared machine therefore never changes its prediction: its stored syndrome is redundant for this readout, and its belief coordinate never leaves its initial level. This matches the registered payoff result. The accuracy-scored comparison between the machine and a syndrome-only decoder came back exactly nil (P5.2, REGISTERED-NEGATIVE). The reason is also exact. REP(3) sees only the bit-flip part of the noise, and with flip probabilities below $1/2$ in both modes the most likely correction is the same in every mode, so accuracy-scored decoders tie by construction.

Memory’s measured value, under a proper scoring rule. Accuracy cannot see the value of memory here, so a post-registration study, labeled as such and leaving every frozen verdict untouched, rescored next-round predictions by negative log-likelihood. This is a strictly proper scoring rule [25], under which better-calibrated probabilities earn credit even when the most likely outcome does not change. At each operating point, one 300,000-round trajectory of REP(3) under N4 is split in half. Anything that is fitted uses only the first half, and every predictor is scored on the same second half. All predictors read the full round outcome: both syndrome bits and the increment of the logical observable. They are therefore predictors of the declared signal, not syndrome-only decoders that could be deployed as such. Figure 5 shows the results, as gains in nats per round over a static predictor that ignores history (MEASURED).

- A *mode oracle*, which is told the hidden mode, gains 0.0109 at the frozen defaults and 0.0582 at a declared louder operating point.
- The *Bayes filter* [58], which knows the model and the initial mode but not the current one, gains 0.00190 and 0.0365.
- A *run-length machine* with K states counts consecutive rounds on which one syndrome bit and the logical increment are both trivial (capped at $K - 1$), and predicts from a table trained on the first half. It gains a positive amount at every K tested. Its gain rises with K at the frozen defaults (0.00015 at $K=2$ to 0.00091 at $K=16$) and up to $K=8$ at the louder point, where $K=16$ sits slightly below $K=8$ (0.01036 vs. 0.01052).
- *Belief rounding*, which snaps the filter’s continuous belief to one of K levels fitted on the first half at every step, loses value at small K . At the frozen defaults it is negative at every tested resolution ($-0.0136, -0.0044, -0.0052, -0.0044$ for $K = 2, 4, 8, 16$; worst at $K=2$ and not monotone in K). At the louder point it is negative up to $K=8$ and positive at $K=16$ ($+0.0281$). That positive value contradicted our stated expectation that rounding would lose value at every tested K , and we report it as such.

Relative to the mode oracle, the Bayes filter recovers about 17% of the oracle’s gain at the frozen defaults and about 63% at the louder point, and the 2-state run-length machine recovers about 8% and 12% of the filter’s gain (INTERPRETATION). The oracle’s measured gains are finite-sample estimates; they lie within 0.0003 of the analytic stationary information values (0.0107 and 0.0579), which are a different quantity. Our packaging-theoretic reading (INTERPRETATION): summarizing a short run-length statistic of the observed record is cheap and safe, whereas quantizing an *internal belief* has a resolution threshold below which it destroys value.

5.4 Pricing: exact machinery, negative evidence (E6)

The pricing audit is this paper’s clean full negative, and we give it the same prominence as the positives. The machinery is exact where declared and runs correctly. On REP(5), exhaustive enumeration gives the value curve $V(b)$ and the shadow prices $\lambda(b)$ as exact rationals (EXACT-FINITE). The greedy selector comes within 0.00405 of the exact curve at every budget (MEASURED). Every registered *prediction* about pricing behaviour failed at the frozen tolerances (REGISTERED-NEGATIVE throughout):

- λ was predicted to be nonincreasing after its maximum. Instead, λ jumps back up at several budgets ($b = 3, 7, 10, 12, 15$), an irregular, nonmonotone pattern that comes from mixing degree-1 checks (cost 1) with degree-2 checks (cost 2).
- A slack point $b^* < b_{\max}$ was predicted. Instead $b^* = b_{\max}$ for both code families (next section).

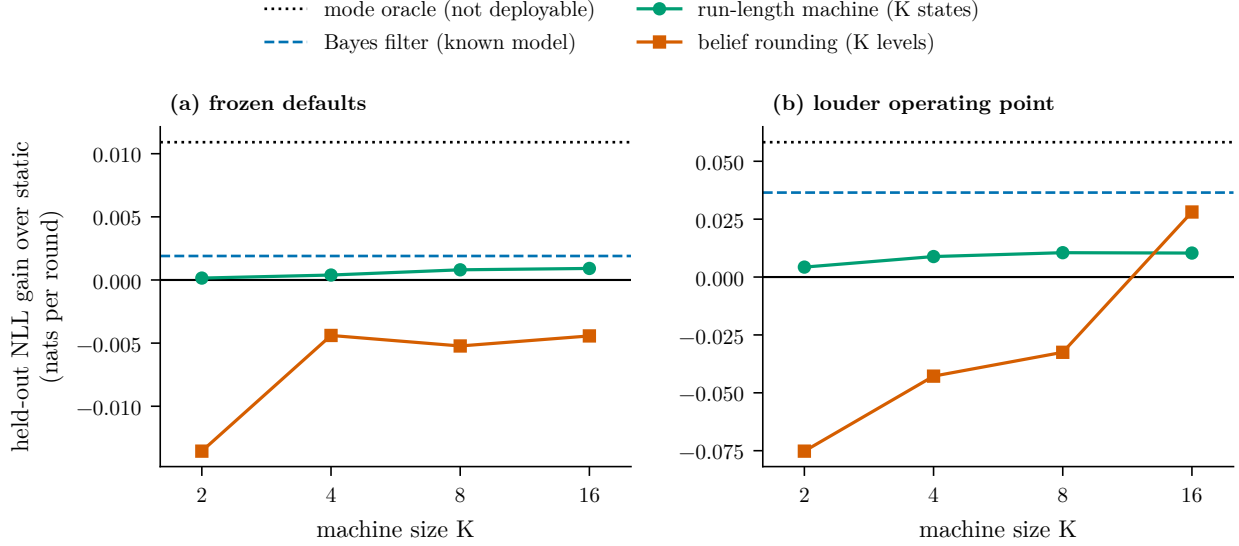


Figure 5. The measured value of decoder memory under a proper scoring rule: held-out gain in next-round log-likelihood over a static predictor, on REP(3) with hidden-mode noise N4, at the frozen defaults (a) and at the declared louder operating point (b) (E5, post-registration study; MEASURED). Horizontal lines: the mode oracle, which is told the hidden mode and is not deployable, and the Bayes filter, which knows the model. Markers: run-length machines and per-step belief rounding at $K = 2, 4, 8, 16$; the lines connecting them only guide the eye. All predictors read the full round outcome, including the logical increment, and all are scored on the same held-out half of each trajectory. Note the different vertical scales of the two panels.

- The consequence test (removing the last unit before b^* should shift the sampled logical error rate detectably) did not separate: $\Delta = -0.000135$ against a combined standard error of 0.000119.
- Permuted proxy costs were strictly worse at only 36.9% of budget points, against a registered bar of 80%.

We do not soften this. As instantiated and thresholded here, the pricing primitive produced no positively evidenced result. Section 6 suggests that the failures are errors of threshold scale and test design rather than refutations of the pricing concept (INTERPRETATION), but on this prototype’s evidence the primitive stands negative.

6 Results II: the failed predictions as a taxonomy

Fourteen of the original build’s 25 registered predictions failed as frozen. Appendix B lists every one with its verdict and artifact. The frozen bars are in the public design documents and the measured values in each experiment’s results file, and diagnosing them required changing no frozen parameter. This section does two things: it sorts the failures into three classes, and it works through one example of each in enough detail for the reader to judge whether our readings are honest. The classification is ours (INTERPRETATION):

1. **Bounds written at the wrong scale.** The prediction constrained the right quantity, but its bar was set at the scale of a different one.
2. **Thresholds frozen too strictly.** The predicted effect exists and points the right way, but the frozen numeric bar demanded more than the effect delivers.

3. Structure the prediction got wrong. The measurement is informative and the machinery correct, but the system did not behave as the registered hypothesis said.

Class 1: a bound at the wrong scale (P1.1). We registered that, at $p = 1/20$, the full-schedule residual trace would fall below 10^{-2} and shrink tenfold from distance 3 to distance 5. Measured: 0.0837 and 0.0740, a ratio of 1.13. The bound was written at the scale of a logical error probability. But $\text{tr}\Xi$ is the mean-squared error of the best affine estimator of a ± 1 observable, a different quantity with a different scale and different dependence on distance. The failure is real and it taught us something concrete: predictions must be written in the object’s own units, and the residual’s units are variance, not failure probability. The limitation of the linear class behind this number is also visible error by error. A deterministic cross-check that enumerates the low-weight errors shows that the affine estimator A^* built from REP(5)’s four syndrome bits agrees with majority-vote (minimum-weight) decoding on 14 of 16 of them and fails on two (exact coefficients, evaluated in floating point). That this reflects a limitation of the whole affine class, not just of this estimator, is our reading (INTERPRETATION).

Class 2: a threshold frozen too strictly (P6.2). We registered that a slack point exists strictly before the largest examined budget, at tolerance $\lambda_{\text{tol}} = 10^{-9}$. Measured: $b^* = b_{\text{max}}$ for both REP(5) and SURF(3). The exact REP(5) curve shows why. Its late marginal values are tiny, of order 10^{-5} to 10^{-4} , and the step from budget 14 to 15 adds exactly nothing. But the last step, from 15 to 16, still adds about 1.7×10^{-5} , far above the tolerance; so the condition “ λ stays below tolerance from here on” first holds at b_{max} . For REP(5), the intended phenomenon, that most of the budget buys almost nothing, is plainly present in the curve: at a tolerance of 10^{-3} the slack point would be $b^* = 4$, and at 10^{-4} it would be 13. SURF(3) is different. Its budget range ends when all eight checks are scheduled, and its greedy curve still gains more than 0.008 at that last step, so even those looser tolerances give $b^* = b_{\text{max}} = 8$. (For SURF(3) the curve is the greedy one, a computed lower bound on the value function.) Our reading (INTERPRETATION): the registered failure is partly a threshold set four to five orders of magnitude below the phenomenon (REP(5)) and partly a family in which substantial gains continue through the last available check, leaving no plateau before the endpoint (SURF(3)). We did not re-threshold after seeing the data, because a moved threshold is exactly what the discipline forbids.

Class 3: structure the prediction got wrong (P5.3). This is the protocol-trap prediction of Section 5.3. We registered that our alternating schedule was an artifact that internalization would dissolve. Internalization preserved all 4 witnesses; the schedule carries genuine predictive content. The machinery did its job, and it is the registered hypothesis that was wrong. The failure also sharpened a concept: “externally scheduled” does not imply “predictively empty.” For this package and its declared observation catalog, the phase carries information about future observables, so the trap did not classify it as an artifact.

The failure that the second half of the paper grows out of. One more failure deserves mention: P2.1, the deployable drift witness that never reached its detection criterion within the tested grid. It belongs to class 3, with a component of implementation defect. Its diagnosis and repair are the subject of Section 7.

7 Results III: a registered failure, diagnosed and repaired inside the framework

This section is the centerpiece of the paper, and Figure 6 is its map. It starts from a registered failure (E2). The failure is diagnosed with the framework’s own exact machinery, and a corrected statistic is re-registered (E7) together with a prediction that it would *still lose* to the baseline on the original test. The correction is then carried to circuit-level noise (E8) and to closed-loop decoder control (E9). Throughout, E2’s committed verdicts stand untouched; everything here is additive.

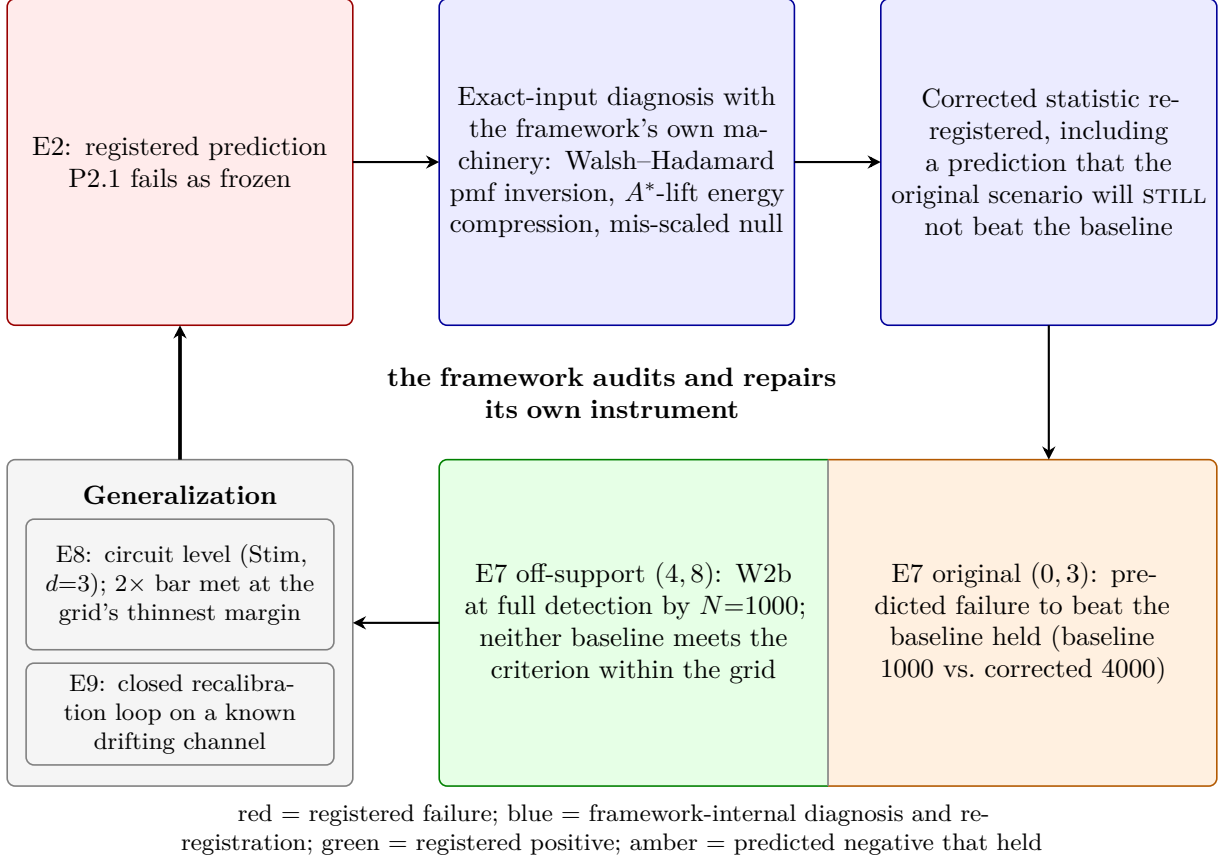


Figure 6. The diagnosis-and-repair cycle of this section. A registered failure (E2, red) is diagnosed with the framework’s own exact machinery (blue). The corrected statistic is re-registered together with a prediction that, on the original scenario, it will *still* not beat the baseline (blue). That predicted negative holds while the off-support scenario detects (amber, green), and the correction then generalizes to circuit level (E8) and to a closed loop (E9). The label inside the loop states the thesis of the arc.

7.1 The failure (E2)

E2 tested drift detection on SURF(3). The decoder’s declared model is N2 with $p = 3/100$. The true model is N3: the same noise plus an unmodeled correlated ZZ error with $q = 1/50$ on the frozen qubit pair (0, 3). Two witnesses were compared with a baseline that monitors the decoder’s logical failure rate. W1 is an *oracle* witness that reads the true logical values in simulation, labeled non-deployable from the start. W2 is the syndrome-only witness, the one that would matter in

practice. A witness “detects” at sample size N when it raises an alarm in at least 9 of 10 seeds. Results (MEASURED): W1 detected at $N = 500$ shots against the baseline’s 1000, a real factor of 2 but short of the registered factor of 4; W2 *never reached the detection criterion within the frozen grid* (up to $N = 16000$; at most 5 of 10 seeds raised an alarm at any N). Both registered predictions failed (REGISTERED-NEGATIVE). The null calibration passed: false-positive rates were 0.0 for W1 and 0.02 for W2 against the registered 2% bar. These are empirical rates over 50 seeded null runs; they estimate, rather than guarantee, the population false-alarm rate.

7.2 The diagnosis: exact inputs, and mostly about the test itself

Before any correction was written, we analyzed the failure with the same moment engine the audits are built on. The mean shifts, covariance blocks and syndrome distributions below are exact rational objects; the scalar energies and divergences computed from them are floating-point evaluations of those exact inputs. Each of the four findings was then pinned as a test in the repository. The reflexive point we draw from findings 1 and 2 is an analogy, marked as such (INTERPRETATION): the moment machinery that audits a schedule’s coverage of the logical observables was turned on the witness itself, to ask how much of the drift signal it kept.

Finding 1: the statistic discards its mean signal (EXACT-FINITE). W2 was built from *centered* covariance blocks, so the drift’s direct effect on the mean entered only through its second-order footprint, never as a mean feature. The exact mean shift caused by the frozen drift is zero on seven of the eight checks and $-331776/9765625 \approx -0.0340$ on the eighth, with total energy $\|\Delta\text{mean}_L\|^2 = 0.0011542$ (a floating-point sum of the exact entries). A real, exactly quantified direct signal was being left out.

Finding 2: a lift built for the wrong objective compresses the signal about $73\times$. W2 mapped the shift in the syndrome covariance through the coverage-optimal map A^* . That map is fitted to predict *logical values*, not to preserve *sensitivity to drift*. The drift signal’s energy is $\|\Delta K_{LL}\|_F^2 = 0.0032632$ before the map and 4.4703×10^{-5} after it (floating-point norms of exact covariance differences). The compression ratio of about $73\times$ is our reading of those two numbers (INTERPRETATION).

Finding 3: a borrowed null (INTERPRETATION, confirmed by measurement below). W2’s alarm threshold was calibrated on W1’s statistic, a differently scaled object, rather than on W2’s own null distribution.

Finding 4: the frozen scenario is structurally hard for syndrome-only detection. This is the key step. Means of XOR-combinations of checks are the Walsh–Hadamard coefficients of the syndrome distribution (Section 3.1), so the full 2^8 -outcome syndrome distribution is exactly recoverable under both the declared and the true model (EXACT-FINITE). From these two exact distributions we evaluate, numerically, the per-shot Kullback–Leibler divergence $D(\text{true} \parallel \text{declared})$ of the syndrome, in nats per shot. It is a per-shot measure of the evidence each shot carries about the drift. By Stein’s lemma [13], it is the best achievable error exponent of an asymmetric test that holds fixed the probability of rejecting the drifted model when the drift is present; holding the false-alarm rate fixed instead calls for the reverse divergence. We use it as an information measure, not as a latency prediction. We compare it with the per-shot divergence of the logical-failure indicator, measured from $N = 2 \times 10^6$ shots per scenario (MEASURED), for three placements of the injected error:

Pair	Relation to logical support	syndrome KL/shot (from exact pmf)	failure KL/shot (measured)	ratio
(0, 3), the frozen E2 pair	2 of 3 qubits of $\bar{Z}$	0.006151	0.009247	0.665
(2, 5)	1 qubit touches $\bar{X}$	0.010823	0.009841	1.100
(4, 8)	off both logical supports	0.051352	0.000066	773.4

Table 3. Per-shot KL divergence (nats per shot) available to the syndrome, evaluated from the exactly recovered distribution, against that of the logical-failure indicator, measured; the ratio is therefore a measured quantity (MEASURED). The frozen E2 pair (0, 3) sits, by construction, nearly on a minimum-weight logical-failure path: the failure channel carries more per-shot information about such drift than the syndrome does (ratio below 1), whereas for the off-support pair the syndrome carries about 773 $\times$ more. Per-shot divergence is a discrimination rate, not a finite-sample latency ratio.

Our structural reading (INTERPRETATION): drift that feeds a minimum-weight failure path is exactly the drift that the failure channel is built to notice and that syndrome decoding is built to correct silently. Of the three scenarios, the E2 pair puts a syndrome-only witness at the greatest disadvantage relative to failure-rate monitoring. This is a fact about this grid of scenarios, not a general worst-case statement, and it holds independently of findings 1–3. It shaped the re-registration: the corrected witness had to be tested across the *range* that finding 4 exposes, with an honest prediction attached to each regime.

7.3 The corrected witness, re-registered with a predicted failure (E7)

The correction addresses findings 1 and 2 with one object. Take the vector of means of the degree- ≤ 2 probe family: all 8 native checks plus all 28 pairwise XOR products, that is, 36 exact Walsh–Hadamard coefficients. Use it *directly*, with no centering and no lift. The corrected detection statistic is the quadratic form

$$T = (\hat{\theta} - \theta_{\text{model}})^\top \Sigma_\theta^+ (\hat{\theta} - \theta_{\text{model}}), \quad (3)$$

where $\hat{\theta}$ is the empirical mean over N shots, and the reference mean θ_{model} and estimator covariance Σ_θ are exact under the declared model. It is a model-referenced Mahalanobis form of Hotelling type [34]; unlike the classical T^2 , it uses the declared model’s covariance rather than a sample estimate, and with the pseudoinverse it is the square of a seminorm. We claim no chi-square null distribution or power theorem for it. Instead, its alarm threshold is calibrated on its own null by a seeded parametric bootstrap, that is, Monte Carlo draws from the declared model [18].

Around T we registered a ladder of four rungs. W2a is the *old* statistic with only its null recalibrated, which isolates finding 3. W2b is the quadratic statistic of Eq. (3). W2c adds a matched-filter *naming* step: it whitens the deviation, matches it against a dictionary of nine exact per-qubit drift signatures, and maps the best match forward to a logical direction. W2d wraps the statistic in a CUSUM sequential test [52], and the baseline receives the *same* sequential upgrade. That removes one asymmetry between the two sides of the test design; it does not equalize their operating characteristics, and the null-rate checks below report those separately. Seven predictions were frozen from finding 4’s numbers before implementation. Three outcomes carry the section (Figure 7).

The off-support scenario: a large registered win. On pair (4, 8), where Table 3 put about 773 $\times$ more per-shot information in the syndrome, the corrected rungs detect at $N = 1000$ (W2b and W2c) and $N = 2000$ (W2d). Neither baseline reaches the detection criterion anywhere in the

frozen grid up to $N = 16000$. The registered bar asked for at least a $10\times$ advantage, and it held (REGISTERED-POSITIVE).

The original scenario: the predicted failure held. Backed by Table 3’s ratio of 0.665, we registered in advance that on the original E2 pair $(0, 3)$ *no corrected rung would beat the baseline*. That is what happened: the baseline detects at $N = 1000$, the best corrected rungs at $N = 4000$ (REGISTERED-POSITIVE, a predicted negative that held). We give this result more weight than the win above, deliberately. A correction developed after a failure invites the suspicion of post-hoc storytelling: a statistic quietly tuned until the old test passes, the degenerating move in Lakatos’s sense [44]. A frozen prediction, backed by exact computation, that the correction would *still lose* on the original test, followed by exactly that loss, is the strongest evidence this arc can offer that the diagnosis explained the failure rather than fitting it. The near-parity pair $(2, 5)$ was registered as descriptive, with no bar. It landed where its ratio of 1.100 suggested: baseline at 1000, W2b and W2c at 2000, W2d at 8000 (MEASURED).

What failed. Three E7 predictions failed (REGISTERED-NEGATIVE each). First, the recalibrated-null prediction P7.1: we predicted that the old statistic’s own-null threshold would be at most 10% of the borrowed one, and measured a ratio of 1.33. Our reading (INTERPRETATION): the reasoning behind the prediction compared the *injected signal’s* scale, of order p against p^2 , rather than the scale of *null fluctuations* that a threshold actually calibrates. Second, naming split (Figure 8). Only 3 of 10 detecting seeds named a qubit in the true injection pair, against a bar of 8 of 10, while the forward-mapped overlap with the true logical direction was 0.869, above its 0.6 bar (MEASURED). Our reading (INTERPRETATION): the naming dictionary is a signed matched filter built from finite differences of the exact signatures, and the overlap is taken in the two-dimensional space of the selected logical observables. The nine per-qubit signatures map to few distinct directions there, so a high overlap is a coarse check: it can pass while qubit-level naming fails, and it does not by itself localize the drift physically. Third, the headline quadratic rung W2b had a null false-positive rate of 6% against the 2% bar. The other separately calibrated statistics, W2a and W2d, and both baselines passed; the naming rung W2c detects with W2b’s statistic and threshold, so it shares W2b’s calibration.

7.4 Generalization to circuit-level noise (E8)

The corrected statistic carries over directly to circuit-level noise. Detectors replace checks, and the degree- ≤ 2 family becomes the detectors plus their pairwise XOR products: 40 detectors and 820 features for the rotated distance-3 Stim memory circuit used. Exact Gaussian elimination over $\mathbb{F}_2$ on the declared detector error patterns gives rank 40 of 40, so for this positive-noise model the detectors span their full space. There is no exact moment engine at circuit level, so θ_{model} and Σ_θ are estimated from a large calibration sample of the declared model (2×10^5 shots), and the predictions were frozen from a disjoint-seed pilot before the scored run (Section 4.1). The scored drift is a global doubling of the error rates. The baseline is the logical failure rate of a PyMatching decoder, under the same detection criterion.

The witness detects at $N_{\text{det}} = 50$ against the baseline’s 100 (Figure 9). This clears the pilot-frozen $2\times$ bar exactly. On the discrete grid $\{10, 20, 50, 100, 250, 500, 1000, 2000\}$ that is the narrowest margin the grid can resolve, and the result should be read with that caveat attached, not as a comfortable win (REGISTERED-POSITIVE, margin caveat mandatory). Null false-positive rates were 0% for both sides (REGISTERED-POSITIVE). One further detail is preserved in the deviations log: the

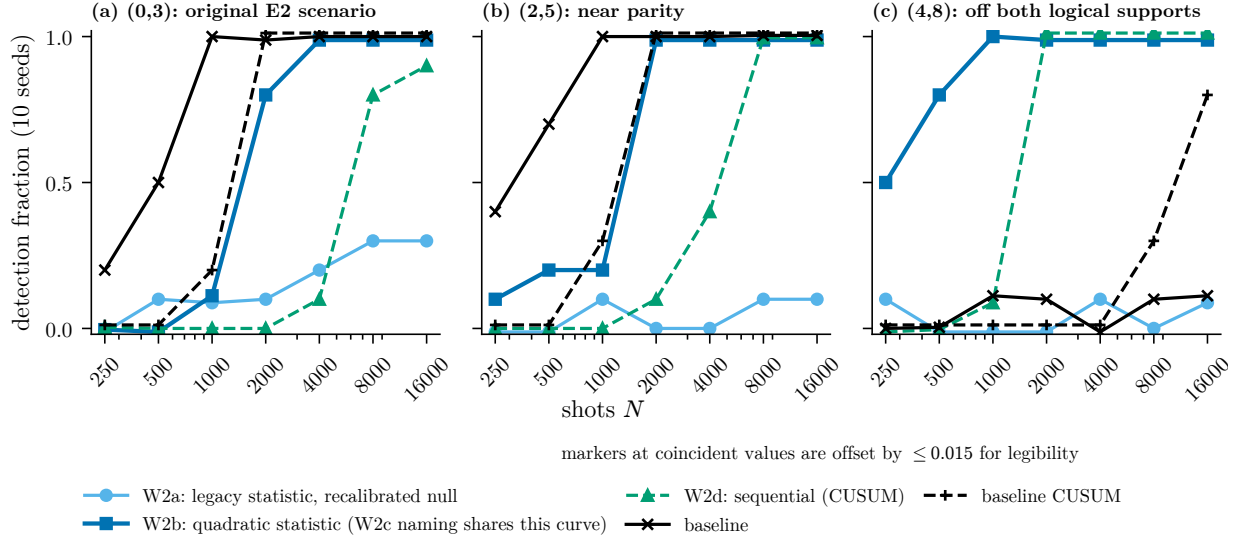


Figure 7. Detection fraction of the corrected witnesses versus shots across the E7 scenario grid, 10 seeds per point (MEASURED). (a) The original E2 pair (0,3): the baseline detects first, which was the registered outcome, not a surprise (P7.3; REGISTERED-POSITIVE as a predicted negative that held). (b) The near-parity pair (2,5), registered as descriptive with no bar. (c) The off-support pair (4,8): the quadratic statistic W2b (whose detection curve the naming rung W2c shares) reaches full detection at $N = 1000$, while neither baseline reaches the 9-of-10 detection criterion anywhere in the grid (REGISTERED-POSITIVE against a registered $10\times$ bar). W2a is the legacy statistic with only its null recalibrated.

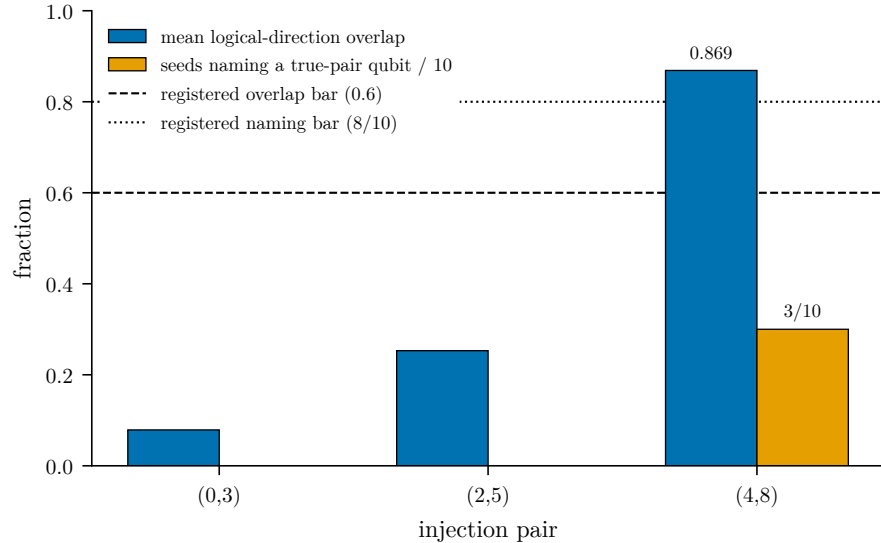


Figure 8. The E7 naming outcome (P7.5; REGISTERED-NEGATIVE). On the off-support pair (4,8) the forward-mapped overlap with the true logical direction is 0.869, above the registered 0.6 bar, while only 3 of 10 detecting seeds named a qubit in the true injection pair, below the 8-of-10 bar. Pairs (0,3) and (2,5) are shown for context; the prediction concerned (4,8). Our reading (INTERPRETATION): the overlap lives in the two-dimensional space of the selected logical observables, where the nine per-qubit signatures map to few distinct directions, so on its own it cannot discriminate between qubits.

pilot had suggested an advantage of about $5\times$, and the scored run delivered $2\times$. The pilot number is a smaller-sample estimate and is not the headline.

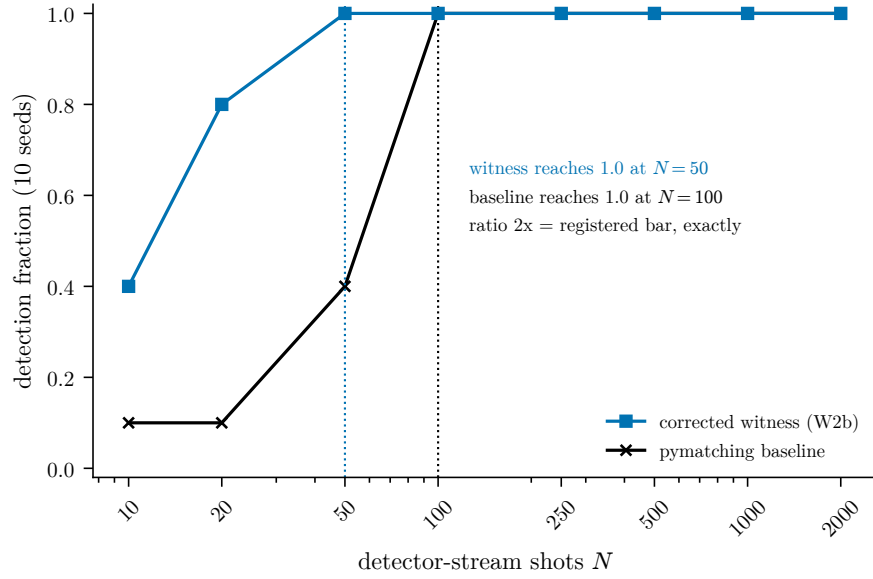


Figure 9. Circuit-level detection fractions under a global doubling of the error rates, on the rotated distance-3 Stim memory circuit, 10 seeds (E8; REGISTERED-POSITIVE). The corrected witness reaches full detection at $N_{\text{det}} = 50$ and the PyMatching baseline at 100. The witness clears the pilot-frozen $2\times$ bar exactly, at the narrowest margin this discrete grid can resolve.

7.5 Closing the loop: witness-triggered decoder recalibration (E9)

The last experiment asks whether the witness’s information is worth anything *operationally*. Suppose the drifting channel is known by construction and the only question is *when* to recalibrate. Does recalibrating when the witness fires buy logical fidelity per unit of recalibration budget?

Setup. Time is divided into 40 epochs of 300 shots each, on a distance-3, 9-round memory circuit (72 detectors, 2628 degree- ≤ 2 features; exact $\mathbb{F}_2$ rank 72 of 72). From epoch 15 on, one measurement-flip channel is elevated $20\times$. We chose a drift of a single channel because, in the pilot, uniformly rescaling all error rates changed the matching decoder’s predictions only very slightly, which would leave a recalibration nothing to show. This is an empirical choice, not an invariance theorem. Uniform rescaling does change the matching weights, and not in proportion: doubling every rate of the distance-3 memory circuits from 0.005 to 0.01 (E8’s drift) multiplies individual edge weights by factors between about 0.76 and 0.89. Recalibrating means matching the observed mean detector-flip rate to a simulated curve of candidate values for the drifting channel, then rebuilding the decoder from the chosen candidate’s detector error model. Five decoder policies run over 10 seeds:

- *static*: never recalibrate (the floor);
- *oracle*: switch to the decoder built for the true model at the true drift epoch, without being charged a recalibration event;
- *budget-matched blind schedule*: recalibrate at uniformly random epochs;
- *frequent blind schedule*: recalibrate every fifth epoch, 8 times per seed;

- *witness-triggered*: recalibrate when the quadratic witness, evaluated on each epoch’s detector batch against a calibrated threshold, fires; then keep monitoring against the recalibrated reference, which corrects false alarms.

Two scope points follow from the design. The witness policy works on whole epochs: when the witness fires on a completed epoch’s detector data, the recalibrated decoder decodes that same epoch. This is legitimate batch decoding with one epoch of latency. The candidate value is chosen from detector statistics, never from logical outcomes. The oracle is a model-known minimum-weight-matching comparator; it is not a proof of an optimal decoder, and it does not bound the measured error rates of other policies pointwise.

How the budget was matched. Experiments like this are easiest to rig by accident at the budget-matching step, so we describe it exactly. On each seed, the matched blind schedule recalibrates at exactly as many epochs as the witness policy did on that seed, chosen uniformly at random on an independently generated timeline. The realized counts are [1, 1, 2, 2, 1, 1, 3, 2, 1, 2] for both policies, mean 1.6. The match is therefore exact in recalibration events and is fixed retrospectively, after the witness has run; it is an offline counterfactual and does not match every computational or calibration cost. The frozen run was first executed with a fixed-period schedule aimed at one event per run, while the witness policy’s realized mean was 1.6. Internal review caught the mismatch, the schedule was changed to exact per-seed matching, and the experiment was re-run and re-frozen with the prediction bars unchanged (Appendix D).

Results. Figure 10 shows the mean post-drift logical error rate per epoch, averaged over seeds.

- Witness-triggered: 0.0930, about 0.0014 above the oracle’s 0.0916 (gap 0.00143 to three significant figures). The frozen bar was a gap of at most 0.01 (REGISTERED-POSITIVE).
- Against the budget-matched blind schedule: 0.0930 vs. 0.1115. At identical per-seed budgets the witness policy is lower by 0.0185 (REGISTERED-POSITIVE). Seed by seed, it is lower on 9 of 10 seeds, and the paired standard error of the mean difference is 0.0035 (MEASURED).
- The static floor sits at 0.1185.
- The frequent schedule spends 8 events per seed, five times the witness policy’s mean, and scores 0.0946. Its deficit of 0.0016 to the witness policy is smaller than the paired standard error of 0.0020 at 10 seeds, so this comparison is unresolved (MEASURED, no bar).
- Pre-drift trigger rates, descriptive: on the scored seeds the witness never triggered before the drift, while the matched and frequent schedules recalibrated before the drift on 50% and 100% of seeds.

Two scope statements bound this result. First, the resolved comparison is *witness against budget-matched blind scheduling*. Whether the witness beats *frequent* blind scheduling is unresolved at this sample size; spending five times the budget blindly closes most, but not all, of the gap. Second, the recalibration searches only the one channel known by construction to drift. E9 therefore demonstrates the value of the witness for *timing and budget*, not for drift *identification*, which is E7’s naming step and was tested separately.

8 Related work

The QEC mechanisms this paper touches are established, and we state the overlaps specifically.

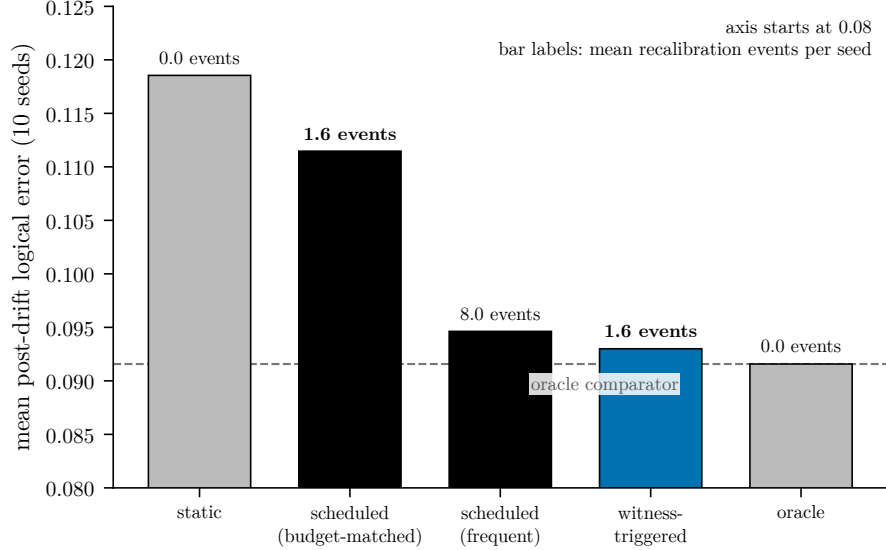


Figure 10. Closed-loop decoder recalibration (E9): mean post-drift logical error rate per epoch over 10 seeds, with each policy’s mean number of recalibration events per seed above its bar (MEASURED). Recalibration re-estimates only the one channel known by construction to drift. The oracle switches to the true-model decoder at the drift epoch without spending a recalibration event, which is why it shows 0.0 events. The witness-triggered policy and the budget-matched blind schedule spend identical event counts on every seed (between 1 and 3, mean 1.6), and the witness policy is lower by 0.0185 (REGISTERED-POSITIVE). The frequent schedule spends 8 events per seed; its 0.0016 deficit to the witness is smaller than the 0.0020 paired standard error at 10 seeds and is unresolved, so the figure does not show that the witness beats frequent blind recalibration.

Noise calibration from detector statistics. Calibrating noise models from pairwise detector correlations, through the p_{ij} estimator and the edge-weight formula $w_{ij} = \log((1 - p_{ij})/p_{ij})$, is standard experimental practice [20, 26], deployed at scale in Google’s surface-code experiments [27, 28]. Detector statistics have likewise been used to detect and remove correlated and leakage errors [47], and hidden-Markov inference on repeated syndrome records has been used to detect leakage [9, 80]. E9’s recalibration is simpler than any of these: it matches the mean detector-flip rate to a simulated curve for the one channel known to drift and rebuilds the decoder from the chosen candidate’s detector error model. Estimating the noise model from syndrome statistics alone, without disturbing the logical information, has its own literature: in-situ estimation proposals [21, 12], identifiability results [81, 82], and learning of detector models and correlated noise from experimental syndrome data [10, 30, 23, 59, 5].

Decoders under drift. Reweighting a matching decoder’s graph in response to drifting noise is the direct subject of prior work [64, 35, 83], as is calibrating decoder priors to correlated or device-specific noise [49, 63]. Among the closest neighbours of Section 7’s drift witness that our bounded literature search found are the time-varying syndrome estimators of [42, 4] and the drift-triggered recalibration of [56]. In-situ recalibration of physical controls during repeated error detection was demonstrated in [38]. Drift itself is a characterized phenomenon [57], and sequential change detection by CUSUM is classical [52].

Budgeted syndrome measurement. Trading measurement budget against decoding quality by measuring fewer, or adaptively chosen, checks is also established [2, 65, 3], though not with a coverage-residual objective.

Tools. Our simulation and decoding infrastructure is Stim [24] and PyMatching [31, 32], used as documented libraries.

What this paper adds. Relative to all of this, we claim no novelty of mechanism. What we add is orthogonal. First, a certificate framing in which coverage, existence, memory and price are typed audit objects derived from one framework. Second, an exact-computation layer, notably the Walsh–Hadamard inversion of the syndrome distribution behind Section 7.2, used after E2 had failed to *explain* why a detector-statistic method failed, and then to predict in advance where its corrected form would and would not win. Third, a pre-registration discipline under which the framework’s predictions, including predictions of its own failure, were frozen and scored.

The SBT side. This paper takes the SBT papers [69, 70, 78, 72, 75, 73, 68, 79, 77, 76, 71, 74] as its specification and tests them; it does not extend the theory. Appendix E cites the instantiated formulas object by object, and Section 2 notes, primitive by primitive, the established external concepts (lumpability, causal states, projection-operator memory) that the instantiated objects correspond to.

9 Discussion: what this does and does not establish

What is established, on this prototype’s evidence. The following is our synthesis of the results (INTERPRETATION). Four SBT primitives, forced into a technical domain, each produced a concrete, machine-checkable audit object that behaved on ground truth as intended. The coverage residual lands exactly on the enumerated minimum mean-squared error where the theory says it must. The existence certificate’s typed statuses do genuine classification work, including refusing a trivialized decoder. The memory witness comes with an exact, rational worst-case gap, and currentization names the hidden coordinate. The protocol-trap control operated correctly against our own hypothesis. The falsification discipline was real: 14 of the original 25 predictions failed as frozen, and none was tuned away. And the discipline was *productive*. The framework’s exact machinery diagnosed one of its own registered failures and predicted in advance both where the corrected instrument would win and where it would keep losing; both predictions held. The correction then carried over to circuit level, clearing its bar at the narrowest margin the grid resolves, and to a closed control loop on a known drifting channel, with the frequent-schedule comparison left unresolved.

What is not established.

- No hardware evidence of any kind; everything is classical simulation of small codes.
- No claim that a nonzero residual rules out a nonlinear decoder.
- No claim that the corrected witness beats failure-rate monitoring in general. The advantage depends on the scenario, and the original E2 scenario is our own registered counterexample.
- No claim that witness-triggering beats *frequent* blind recalibration; that comparison is unresolved at this sample size. The resolved comparison is budget-matched.
- No positive evidence for the pricing primitive, whose registered predictions all failed, and no evidence at all about transport, which was never built.

- No Born-rule, Bell or coherent-dynamics claims, and no asymptotic-threshold claims: every result concerns a finite carrier in classical simulation.
- No certificate rests on a fitted model; fitted quantities appear only as diagnostics or as measured payoffs.
- No inference across audits: stability does not establish nontrivial storage, coverage and existence do not imply one another, and memory witnesses and drift do not imply one another.
- No claim of drift identification from E9: its recalibration acts on the one channel known by construction to drift, so it evidences the timing and budget value of the witness only.
- Nothing here validates SBT’s broader ambitions in other domains. One domain, carefully instrumented, is one data point.

What the exercise suggests about frameworks for emergence. The pattern this prototype shows is: primitives that compile to computable certificates, nulls that can fire against the theory’s favourite conclusions, predictions frozen before the scored runs, and a repair loop that runs *inside* the framework. We think this is the right evidentiary standard for any theory of emergence that wants to be more than vocabulary. Quantitative accounts of emergence outside SBT likewise tie a macro-level description to a computable criterion: coarse-grained predictability [36], effective information [33], dynamical independence [1]. That they share the demand for falsifiability is our reading (INTERPRETATION), not a claim those works make. Judged by that standard, four of SBT’s six primitives produced positively evidenced objects here, one (pricing) returned a clean registered negative, which is the discipline working rather than failing, and one was not attempted. Whether the primitives keep earning their place is now an empirical question with a template: pick a domain with a packaging structure, instantiate, register, run, and report the failures as fully as the successes.

10 Reproducibility statement

Everything reported above regenerates from the public repository. The reported numbers were produced in this environment: Python 3.12.3, numpy 2.5.2, scipy 1.18.1, stim 1.16.0, pymatching 2.4.0, matplotlib 3.11.1 and pytest 9.1.1, with single-threaded linear algebra (OPENBLAS_NUM_THREADS=1, OMP_NUM_THREADS=1). Each experiment directory records its environment. The dependency floors in `pyproject.toml` are lower bounds, not pins; to reproduce the exact environment, install these versions explicitly. Byte-identity is a property of one fixed environment: across library versions or thread counts, floating-point fields can differ in their final bits even when every reported value and verdict agrees. Setup and verification:

```
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
pip install -e ".[dev]"           # package + pytest
pytest -q                        # 209 tests pass
python -m sbqos.reproduce_all       # nine experiments; manifests verified
python -m sbqos.reproduce_all       # second run; compare byte-for-byte
python paper/figures/make_figures.py # figures F3-F9 (deterministic)
```

The suite has 188 tests of the prototype and 21 of the paper’s figure pipeline. The tests that check the theorem anchors are

- `test_thm_chain_rule_rep5_seeded_splits_exact,`
- `test_thm_top_ladder_rep3_mmse_matches_bruteforce_exact,`
- `test_thm_B5_lumpable_zero_deficit_for_syndrome_lens.`

Each `reproduce_all` run regenerates the nine experiments and verifies the manifest it has just written. The evidence recorded for this paper is stronger: consecutive runs produced byte-identical files, and a fresh run from a clean export of the repository reproduced the committed manifests byte for byte, in about 8 minutes on the reference machine. Every experiment directory contains its frozen configuration, results, SHA-256 manifest, environment record and the exact data behind every figure, as CSV. The manifests are committed to the repository, so a reader can confirm that they regenerated the same evidence, not merely similar evidence.

A Notation

Symbols used in the paper, grouped by primitive. The repository’s notation file maps each symbol to its name in the code.

Symbol	Meaning
<i>Packaging (Section 2)</i>	
Z, P, f	carrier state set, one-step Markov kernel, lens (macro label map)
μ	a carrier distribution (the argument of the packaging operator)
u_x	prototype distribution of macro label x
Q_f, U_f	marginalization through the lens; lift of a macro label to its prototype
$E_{\tau, f} = U_f \circ Q_f \circ P^\tau$	packaging operator at horizon τ
π	stationary law of P (a declared finite-horizon weighting replaces it for absorbing models)
σ_a	± 1 observable of Pauli vector a : $\sigma_a(e) = (-1)^{\langle a, e \rangle}$
L, D	native (scheduled) and dissolving (logical) probe families
<i>Coverage audit</i>	
K_{LL}, K_{DL}, K_{DD}	covariance blocks of scheduled-check / logical ± 1 observables
$\Xi(D \mid L)$	adequacy residual: residual covariance of the best affine estimator of D from L
A^*	coverage-optimal linear map, $A^* = K_{DL} K_{LL}^+$
$\lambda_{\max}(\Xi - \Omega), z$	blind-spot witness eigenvalue / direction; Ω = calibrated null allowance
$L^{(2)}$	degree- ≤ 2 family: native checks plus all pairwise XOR products
$\hat{\theta}, \theta_{\text{model}}, \Sigma_\theta$	empirical / exact mean vector of $L^{(2)}$ and its estimator covariance
T	corrected quadratic drift statistic, Eq. (3)
N_{det}	smallest sample size with detection in $\geq 9/10$ seeds
<i>Existence audit</i>	
δ, ε	idempotence defect; retention error (theorem: $\delta \leq \varepsilon$)
CD_τ	closure deficit, $I(X_t; Y_{t+\tau} \mid Y_t)$
RM_τ	route mismatch: spread of the micro states' coarse transition profiles within each macro cell
Δ_{pred}	held-out gain in next-label log-likelihood from adding the previous macro label (a stream proxy compared with CD)
mult.	multiplicity: number of stable logical prototypes (the symbol μ is reserved for carrier distributions)
<i>Memory audit</i>	
Q, M	current / predictive quotients (partitions of the declared history catalog); $ Q , M $ their class counts
MaxFiber	largest number of predictive classes inside one current class
$\Delta^{\max}$	exact worst-case predictive gap (rational); a predictor using only current information errs by $\geq \Delta^{\max}/2$
<i>Pricing audit</i>	
$V(b), \lambda(b), b^*$	value curve over budget; shadow price $V(b+1) - V(b)$ (a finite difference); slack point
<i>Circuit level / closed loop</i>	
p_0, p_1	declared / drifted error probability
d, R	code distance, circuit rounds

Table 4. Notation.

B The complete prediction ledger

Every registered prediction, its verdict against its frozen bar, and its artifact. The original build (E1–E6) has 25 predictions, of which 14 are REGISTERED-NEGATIVE. The follow-up (E7–E9) has 9

scored predictions and 5 declared descriptive reports. Figure 1 shows the same ledger graphically. Artifact paths abbreviate `artifacts/eN_*/.../results.json` as `eN`.

id	short statement	verdict	artifact
P1.1	REP full-check residual bound and distance suppression	REGISTERED-NEGATIVE	e1
P1.2	single-check ablations strictly increase residual; REP(5) end/middle distinction	REGISTERED-POSITIVE	e1
P1.3	witness overlap threshold and directional naming	REGISTERED-NEGATIVE	e1
P1.4	duplicated-check saturation gives zero discharge	REGISTERED-POSITIVE	e1
P2.1	witnesses detect earlier than matched baseline	REGISTERED-NEGATIVE	e2
P2.2	witness direction names the injected logical direction	REGISTERED-NEGATIVE	e2
P2.3	null false-positive rate at $N=4000$ at most 2%	REGISTERED-POSITIVE	e2
P2.4	N5 drift-witness trace Spearman $\rho \geq 0.9$	REGISTERED-NEGATIVE	e2
P3.1	greedy residual no worse than tested baselines at every budget	REGISTERED-POSITIVE	e3
P3.2	degree ladder monotone; top rung equals the exact (conditional-mean) MMSE	REGISTERED-POSITIVE	e3
P3.3	duplicate candidates have zero marginal value	REGISTERED-POSITIVE	e3
P3.4	greedy stops before exhausting duplicate-containing list	REGISTERED-POSITIVE	e3
P4.1	baseline CD near zero, multiplicity 2, certified, δ monotone	REGISTERED-NEGATIVE	e4
P4.2	N4 non-closed while per-mode models have zero deficit	REGISTERED-NEGATIVE	e4
P4.3	N5 finite-horizon $RM \geq 10\times$ baseline	REGISTERED-NEGATIVE	e4
P4.4	broken-decoder trivialization guard	REGISTERED-POSITIVE	e4
P4.5	Pearson $r(CD, \Delta_{\text{pred}}) \geq 0.9$	REGISTERED-POSITIVE	e4
P5.1	memoryless package: no witnesses, no payoff gap	REGISTERED-POSITIVE	e5
P5.2	N4 witness and MaxFiber nontrivial with positive bounded payoff	REGISTERED-NEGATIVE	e5
P5.3	protocol-trap witnesses vanish after internalization	REGISTERED-NEGATIVE	e5
P5.4	mode-bit currentization passes at cardinality 1; pure checks do not	REGISTERED-POSITIVE	e5
P6.1	λ nonincreasing after its maximum	REGISTERED-NEGATIVE	e6
P6.2	slack point before the examined budget maximum	REGISTERED-NEGATIVE	e6
P6.3	consequence test separates b^*-1 from b^* by > 2 SE	REGISTERED-NEGATIVE	e6
P6.4	proxy costs strictly worse at $\geq 80\%$ of budget points	REGISTERED-NEGATIVE	e6
P7.1	recalibrated own-null threshold $\leq 10\%$ of borrowed threshold	REGISTERED-NEGATIVE	e7
P7.2	off-support (4, 8): best corrected rung $\geq 10\times$ earlier than baseline	REGISTERED-POSITIVE	e7
P7.3	original (0, 3): <i>predicted</i> that no corrected rung beats the baseline's N_{det}	REGISTERED-POSITIVE	e7
P7.4	near-parity (2, 5): descriptive, no bar	MEASURED	e7
P7.5	off-support naming: $\geq 8/10$ seeds name a true-pair qubit, overlap ≥ 0.6	REGISTERED-NEGATIVE	e7
P7.6	null false-positive rate $\leq 2\%$ for every rung	REGISTERED-NEGATIVE	e7
P7.7	sequential vs. fixed-window reported symmetrically; no bar	MEASURED	e7
P8.1	circuit-level witness detects $\geq 2\times$ earlier than baseline	REGISTERED-POSITIVE	e8
P8.2	circuit-level null false-positive rate $\leq 2\%$	REGISTERED-POSITIVE	e8
P8.cost	cost/scaling report; descriptive, no bar	MEASURED	e8
P9.1	witness-triggered control within 0.01 of oracle	REGISTERED-POSITIVE	e9
P9.2	witness beats per-seed budget-matched blind schedule	REGISTERED-POSITIVE	e9
P9.3	high-budget schedule: descriptive, no bar	MEASURED	e9
P9.4	pre-drift trigger rates: descriptive, no bar	MEASURED	e9

Table 5. The complete prediction ledger. Verdicts are against bars frozen before implementation (E1–E7) or frozen from a disjoint-seed pilot before the scored run (E8–E9).

C Null-battery ledger

The nine standing controls and their outcomes.

1. *Protocol trap / schedule internalization* (E5c). Implemented. Outcome is a negative result for the artifact hypothesis, not a positive trap catch: the registered expectation was that internalization would dissolve the frozen schedule’s witnesses; instead internalization preserves all 4 witnesses and the package classifies `genuine_memory_after_internalization`. The control mechanism ran correctly; it is the “this is an artifact” prediction that failed as registered.
2. *Same-family saturation* (E1 P1.4, E3 P3.3, unit tests). Duplicate-check discharge is exactly zero; duplicate candidates have zero marginal value when passed over.
3. *Null-model calibration* (E2 P2.3). W1 false-positive rate 0.0, W2 0.02 at $N=4000$; REGISTERED-POSITIVE. The follow-up recalibrated every new rung on its own null (E7 P7.6, E8 P8.2), with the headline quadratic rung W2b failing its 2% bar at 6% (REGISTERED-NEGATIVE) as reported in Section 7.3.
4. *Proxy failure* (E6 P6.4). Implemented; proxy strict-worse fraction 0.369 against the registered 0.8 bar—failed honestly.
5. *Slack regime* (E6 P6.2/P6.3, E3 P3.4). E3’s selector stops before duplicate exhaustion; E6’s frozen tolerance was too strict and $b^* = b_{\max}$.
6. *No silent caps* (E3 SURF(5), currentization). SURF(5) logs 62 kept adjacent pairs and 214 dropped out of 276 total degree-2 pairs; the currentization candidate list is declared and bounded.
7. *Trivialization guard* (E4 P4.4). The broken decoder yields $\delta = 0$, multiplicity 1, status `trivialized`; REGISTERED-POSITIVE.
8. *Same-support discipline* (E5, quotient machinery; E7’s scenario grid). Declared history catalogs and scenario grids, no post-hoc edits.
9. *Raw-history comparator* (E5 payoff). A seeded lookup predictor on the last three syndromes (current included), trained on the first half of each trajectory and scored on the second, does not reproduce a positive machine gap—consistently, since the accuracy-scored machine gap is itself 0.0 at frozen defaults.

D Deviations log (summary)

The repository’s deviations file records every divergence between the frozen designs and the running code. It has 11 dated entries, summarized here in a few sentences each; dates and full reasoning are in the file. No entry changed a frozen bar.

Build phase.

1. *Finite-horizon route mismatch for N5* (E4). The latching model is absorbing, so the stationary-weight form of *RM* degenerates. A declared finite-horizon variant was used, and the guard is documented.
2. *Length of the predictive-gap stream* (E4). The fixed length of the proxy’s stream can wash out fast-absorbing finite-horizon effects. This is noted as a caveat on Δ_{pred} , not silently retuned.
3. *Packaged belief automaton* (E5). The minimal decoder machine is built on a declared, quantized packaged-belief automaton rather than on transport checks of linear kernels. No general impossibility of closed finite catalogs is assumed (entry 10).

4. *Bounded-weight cross-check estimator* (E1). The cross-check originally used an uncentered estimator instead of the affine A^* that the decoder actually is. It was corrected before any follow-up work began, giving 4/4 on REP(3) and 14/16 on REP(5), with two genuine failures of the affine class. This entry was added after an external review found the fix documented in prose but missing from the log.
5. *Payoff study* (E5). A post-registration study of next-round log-likelihood, labeled MEASURED, which alters no frozen verdict. An addendum records that its run-length machines read one syndrome bit and the logical increment rather than syndrome history alone.
6. *Granularity of E7's null calibration*. The own-null thresholds of W2a and W2b were calibrated once per sample size N rather than once per pair of N and seed. This was declared; the baseline's calibration was unaffected.
7. *E8 scope and pilot-then-freeze*. Distance 3 only, one global drift scenario, the full pair family, no sequential rung. The pilot seed range and the resulting frozen bars are logged verbatim.
8. *E9 corrections, before and after the first freeze*. The drift scenario was changed from a uniform rescaling to a single elevated channel, the timeline was divided into epochs, and the witness policy needed two pilot-driven fixes: a design that could permanently blind itself after a false alarm, then a noisy-reference bug in the self-correcting version. After the first frozen run, the budget-matched schedule was corrected to exact per-seed event matching, and the experiment was re-run and re-frozen.

Audit of the mathematics and executable certificates.

9. *Exact arithmetic and closure domains*. Exhaustive pricing enumeration, greedy selection, the chain-rule discrepancy, E1's saturation check and E3's MMSE comparison are carried out in rational arithmetic throughout, and uniform stationarity is checked exactly. Closure computations handle arbitrary label sets and skip zero-weight cells, and package construction rejects invalid probability data. Frozen orders and verdicts are unaffected.
10. *E5 sensor, scoring and quotient premises*. The belief update reads the syndrome actually supplied to it: the logical bit is marginalized out, successive cumulative syndromes are differenced, and the known initial state is not treated as an observation. The payoff study fits its rounding grid on the training half and scores every predictor on the held-out half. Hidden-model future catalogs include identity continuations with current tests, so that predictive classes refine current classes (N5 has 8 predictive classes; its witnesses, MaxFiber and $\Delta^{\max}$ are as reported). The machine is treated as a quantized predictor and minimized exactly for its next-syndrome predictions.
11. *E9's uniform-drift rationale*. The single-channel drift rests on the pilot's observation of only a very small decoder difference under uniform drift, not on an invariance argument: uniform rescaling changes the matching weights, and not in proportion. The scenario and the frozen bars are unaffected.

E Formula summary

The instantiated objects, with their SBT sources. All formulas below are as implemented; the design documents in the repository are the normative statement.

Observables and moments [78, 70]. For $a, e \in \mathbb{F}_2^{2n}$ with symplectic pairing $\langle a, e \rangle$: $\sigma_a(e) = (-1)^{\langle a, e \rangle}$, with $\sigma_a \sigma_b = \sigma_{a \oplus b}$. For independent Pauli noise, $\mathbb{E}[\sigma_a] = \prod_i m_i(a)$ with per-qubit rational factors m_i ; all covariance blocks follow in closed form. Means of XOR-combinations of a check family are the Walsh–Hadamard coefficients of its joint syndrome pmf, which is therefore exactly recoverable by inversion.

Adequacy residual and chain rule [78]. $\Xi(D \mid L) = K_{DD} - K_{DL}K_{LL}^+K_{LD}$; $A^* = K_{DL}K_{LL}^+$;

$$\begin{aligned} \text{discharge}(M \mid L, D) &= K_{DM|L} (K_{MM|L})^+ K_{DM|L}^\top, \\ \Xi(D \mid L \cup M) &= \Xi(D \mid L) - \text{discharge}(M \mid L, D), \end{aligned}$$

with the conditional blocks $K_{\cdot|L}$ the usual Schur complements. $\Xi(D \mid L)$ is the residual covariance of the best affine estimator of D from L ; joint covariances are Gram matrices, so K_{LD} lies in the range of K_{LL} and the singular Schur complement is well defined. Blind-spot witness: top eigenpair of $\Xi - \Omega$. Saturation: discharge = 0 exactly for a duplicated probe.

Existence certificate [69, 70, 79, 73, 68]. $E_{\tau,f} = U_f \circ Q_f \circ P^\tau$; $\delta = \max_z \frac{1}{2} \|(E^2 - E)(\delta_z)\|_1$; $\varepsilon = \max_x \|Q_f(u_x P^\tau) - \delta_x\|_{TV}$; theorem $\delta \leq \varepsilon$. $CD_\tau = I(X_t; Y_{t+\tau} \mid Y_t) = \sum_x \pi(x) D_{\text{KL}}(p_x^{(\tau)} \parallel \bar{p}_{\Pi(x)}^{(\tau)})$ under the stationary law π , with $Y = \Pi(X)$ and $\bar{p}_y^{(\tau)}$ the $\pi(\cdot \mid y)$ -weighted fiber average; a declared finite-horizon weighting replaces π for absorbing models (Appendix D); states of zero weight are skipped, so $CD_\tau = 0$ asserts equal coarse rows on the support of the weighting law. $RM_\tau = \sum_x \sum_{z \in B_x} \pi(z) \|R_\tau(z, \cdot) - K_\tau(x, \cdot)\|_1$, with $R_\tau(z, \cdot)$ the coarse transition profile of micro state z , B_x the fiber of macro label x , and $K_\tau(x, \cdot) = \sum_{z \in B_x} \pi(z \mid x) R_\tau(z, \cdot)$ its fiber average (global stationary weight outside, fiber-renormalized weight inside; the same finite-horizon replacement applies to absorbing models). For stationary certificates, status $\in \{\text{certified}, \text{degrading}, \text{non_closed}, \text{trivialized}\}$ by frozen thresholds with first-match priority $\text{trivialized} \succ \text{non_closed} \succ \text{degrading} \succ \text{certified}$. For the absorbing model N5 the stationary certificate is undefined; only finite-horizon diagnostics are computed, and it is reported as `finite_horizon_only` rather than classified.

Predictive quotients [72]. Histories partitioned by exact equality of current signatures (probabilities of all declared current events) and of future signatures (probabilities of all declared (continuation, later-event) pairs, for the hidden-model packages N4 and N5 including the identity continuation with the current events, which makes M refine Q ; for the other catalogs refinement is verified directly) give Q and M ; witnesses are pairs equal in the former, distinct in the latter; $\Delta^{\max}$ is the largest entrywise future-observation difference over witness pairs (exact rational), and any predictor that uses only current information (one prediction per current class) has uniform error at least $\Delta^{\max}/2$. Currentization: smallest declared candidate set whose addition gives witness count 0 and MaxFiber 1. Protocol trap [75]: internalize the schedule as an autonomous state coordinate (random-scan lift) and recompute.

Pricing [77]. $V(b) = \max_{S: \text{cost}(S) \leq b} [\text{tr } \Xi(D \mid L_0) - \text{tr } \Xi(D \mid L_0 \cup S)]$ (exact, by exhaustive enumeration in rational arithmetic, for REP(5); SURF(3) reports the greedy curve, a lower bound on V); V is nondecreasing; $\lambda(b) = V(b+1) - V(b)$ is a discrete finite difference (not a continuous multiplier), and V need not be concave; slack point $b^* = \text{smallest } b \text{ with } \lambda \leq \lambda_{\text{tol}} \text{ thereafter}$; proxy null = seeded cost permutation.

Corrected drift witness (Section 7.3). $T = (\hat{\theta} - \theta_{\text{model}})^\top \Sigma_\theta^+ (\hat{\theta} - \theta_{\text{model}})$ over the degree- ≤ 2 family, with $\hat{\theta}$ the empirical mean of the feature vector F over N shots and $\Sigma_\theta = \text{Cov}(F)/N$ under the declared model (the square of a pseudoinverse seminorm; threshold from a seeded parametric bootstrap of the declared model); naming by whitened matched filter against nine exact per-qubit drift signatures; sequential wrapper $g_t = \max(0, g_{t-1} + \text{increment}_t)$ with null-simulated threshold [52], applied identically to witness and baseline.

F Version history

Version 1 (5 September 2026). First public release, doi:10.5281/zenodo.22337886.

Version 2 (3 October 2026). This version follows an audit of the mathematics and executable certificates behind the paper (the repository’s `design/MATHEMATICS_AUDIT.md`); the corresponding entries are 9–11 of the deviations log (Appendix D). No registered prediction, frozen bar or verdict changed, and the main results stand as in version 1. The changes are:

- The E5 payoff study now fits on the first half of each trajectory and scores every predictor on the same held-out half, with a corrected belief update. The reported gains changed accordingly (Section 5.3; for example, the Bayes filter now recovers about 17% rather than 18% of the oracle’s gain at the frozen defaults, and the worst small- K rounding score is at $K=2$ rather than $K=4$). The qualitative findings are unchanged.
- E5’s predictive quotients now provably refine its current quotients (N5 has 8 predictive classes rather than 5); the operational meaning of $\Delta^{\max}$ is stated as a lower bound of $\Delta^{\max}/2$ on the worst-case error of any predictor that uses only current information; and the minimal machine is described with its exact scope.
- Several statements are given their exact scope: the residual as an affine-estimator residual; closure and route mismatch as statements on the support of the weighting law; the E3 ladder result as squared-error estimation; the E4 correlation proxy; the E9 oracle, batch decoding and offline budget matching; and the reason for E9’s single-channel drift.
- Pricing, greedy selection and the chain-rule discrepancy are computed in rational arithmetic throughout; the test suite has 209 tests; artifacts were regenerated in the environment of Section 10.
- The text was rewritten for readability, with a new abstract; Figure 1 is new and Figure 5 was redrawn.

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